CET PCB 2024 — Paper 1 — v10

Left: book scan. Right: our JSON rendered with KaTeX (the same renderer a website would use). 200 questions.

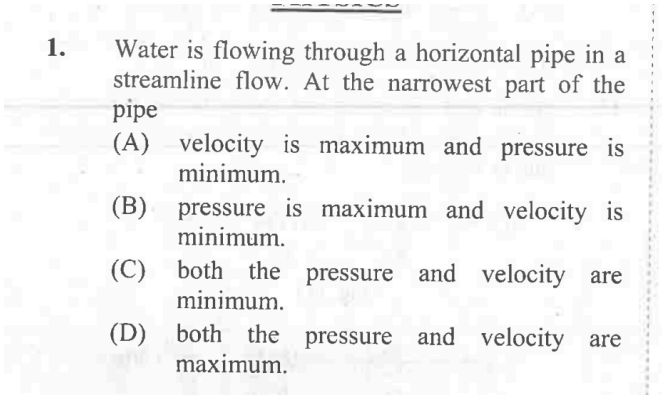
Changes from the previous version (29 fields)

Q	Field	Before	After
Bio_q001	stem	In eukaryotic organisms, replication of DNA occurs in	In eukaryotic organisms, replication of DNA occurs in _____.
Bio_q002	stem	Capillarity theory was proposed by	Capillarity theory was proposed by _____.
Bio_q007	stem	When the phage DNA enters the host bacterium, the host cell protects itself from the viral DNA attacks with the help of enzyme.	When the phage DNA enters the host bacterium, the host cell protects itself from the viral DNA attacks with the help of _____ enzyme.
Bio_q013	stem	DDT resistant mosquitoes and different beak sizes of African seed cracker finches show type of selection respectively.	DDT resistant mosquitoes and different beak sizes of African seed cracker finches show _____ type of selection respectively.
Bio_q015	stem	During oogenesis the meiotic division is arrested at before fertilization.	During oogenesis the meiotic division is arrested at _____ before fertilization.
Bio_q018	stem	Agarose gel electrophoresis is used to separate DNA fragments. The DNA fragments separate due to	Agarose gel electrophoresis is used to separate DNA fragments. The DNA fragments separate due to _____.
Bio_q038	stem	The size of mRNA is generally related to	The size of mRNA is generally related to _____.
Bio_q041	stem	The gene from Bacillus thuringiensis produces a protein that forms crystalline inclusions in bacterial spores.	The _____ gene from Bacillus thuringiensis produces a protein that forms crystalline inclusions in bacterial spores.
Bio_q042	stem	Lateral sulcus separates	Lateral sulcus separates _____.
Bio_q044	stem	Following are viral diseases in poultry, EXCEPT	Following are viral diseases in poultry, EXCEPT _____.
Bio_q045	stem	All the 64 codons in the dictionary of genetic code were deciphered by	All the 64 codons in the dictionary of genetic code were deciphered by _____.
Bio_q046	stem	Median vertical depression of external genitalia in females, enclosing the urethral and vaginal opening is	Median vertical depression of external genitalia in females, enclosing the urethral and vaginal opening is _____.
Bio_q047	stem	Bird pollinated flowers are usually	Bird pollinated flowers are usually _____.
Bio_q048	stem	Complete the following analogy. Parietal wall of Bowman's capsule: squamous epithelium::PCT: epithelium.	Complete the following analogy. Parietal wall of Bowman's capsule: squamous epithelium :: PCT: _____ epithelium.
Bio_q050	stem	Complete the following analogy with respect to classification of mammals. Marsupials: (i): monotremes: (ii) (i) (ii)	Complete the following analogy with respect to classification of mammals. Marsupials: (i) :: monotremes: (ii) (i) (ii)
Bio_q051	option B	tobacco, soybean	tobacco, soyabean
Bio_q051	option C	tomato, soybean	tomato, soyabean
Bio_q066	stem	The number of amino groups in the amino acids present in amides is	The number of amino groups in the amino acids present in amides is _____.
Bio_q069	stem	Hypersecretion of glucocorticoid leads to	Hypersecretion of glucocorticoid leads to _____.
Bio_q070	stem	The restriction enzymes used as molecular scissors are type of	The restriction enzymes used as molecular scissors are type of _____
Bio_q072	stem	Phenotypically tall plants can be obtained from genetically dwarf maize plants by application of	Phenotypically tall plants can be obtained from genetically dwarf maize plants by application of _____.
Bio_q073	stem	Islets of Langerhans are	Islets of Langerhans are _____.
Bio_q074	stem	Homo sapiens are most closely related to	Homo sapiens are most closely related to _____.
Bio_q076	stem	In a male gametophyte, ploidy of tube cell and a male gamete is	In a male gametophyte, ploidy of tube cell and a male gamete is _____.
Bio_q078	stem	The only 5C intermediate formed during reactions of TCA cycle is	The only 5C intermediate formed during reactions of TCA cycle is _____.
Bio_q084	stem	The water which percolates deep in the soil is	The water which percolates deep in the soil is _____.
Bio_q090	stem	Hormone causing vigorous contraction of myometrium to initiate parturition is	Hormone causing vigorous contraction of myometrium to initiate parturition is _____.
Bio_q093	stem	Blood plasma transports oxygen in dissolved state, percentage of which is	Blood plasma transports oxygen in dissolved state, percentage of which is _____.

Bio_q099	stem	Respiratory centres that control the rate and depth of breathing are located in	Respiratory centres that control the rate and depth of breathing are located in _____.

All questions

Phy_q001 key **A**



1. Water is flowing through a horizontal pipe in a streamline flow. At the narrowest part of the pipe
   (A) velocity is maximum and pressure is minimum.
   (B) pressure is maximum and velocity is minimum.
   (C) both the pressure and velocity are minimum.
   (D) both the pressure and velocity are maximum.

Water is flowing through a horizontal pipe in a streamline flow. At the narrowest part of the pipe
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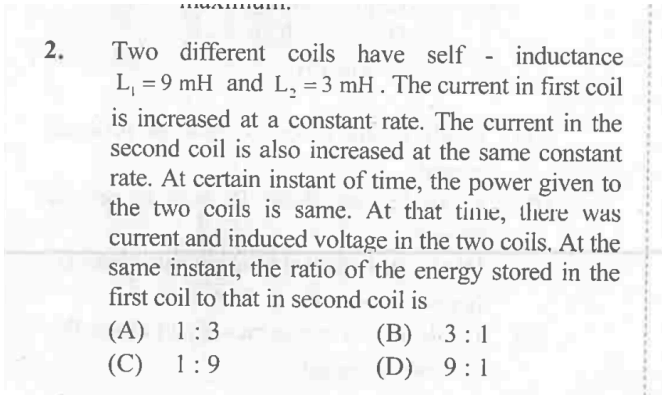
Phy_q002 key **A**



2. Two different coils have self - inductance $L_1 = 9$ mH and $L_2 = 3$ mH. The current in first coil is increased at a constant rate. The current in the second coil is also increased at the same constant rate. At certain instant of time, the power given to the two coils is same. At that time, there was current and induced voltage in the two coils. At the same instant, the ratio of the energy stored in the first coil to that in second coil is
   (A) 1 : 3 (B) 3 : 1
   (C) 1 : 9 (D) 9 : 1

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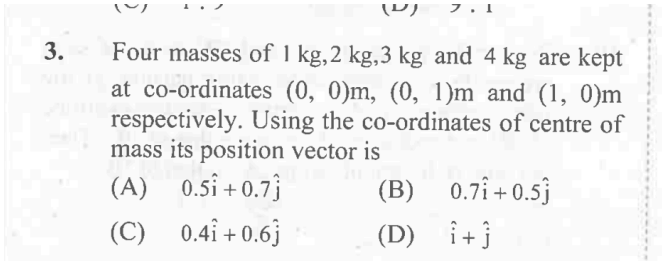
(A) $1 : 3$
(B) $3 : 1$
(C) $1 : 9$
(D) $9 : 1$

Phy_q003 key **B** V9_INVALID

Only three positions are given for four masses (crop shows (0,0), (0,1), (1,0)); centre of mass cannot be computed.



3. Four masses of 1 kg, 2 kg, 3 kg and 4 kg are kept at co-ordinates (0, 0)m, (0, 1)m and (1, 0)m respectively. Using the co-ordinates of centre of mass its position vector is
   (A) $0.5\hat{i} + 0.7\hat{j}$ (B) $0.7\hat{i} + 0.5\hat{j}$
   (C) $0.4\hat{i} + 0.6\hat{j}$ (D) $\hat{i} + \hat{j}$

Four masses of $1\,\text{kg}, 2\,\text{kg}, 3\,\text{kg}$ and $4\,\text{kg}$ are kept at co-ordinates $(0, 0)m$, $(0, 1)m$ and $(1, 0)m$ respectively. Using the co-ordinates of centre of mass its position vector is
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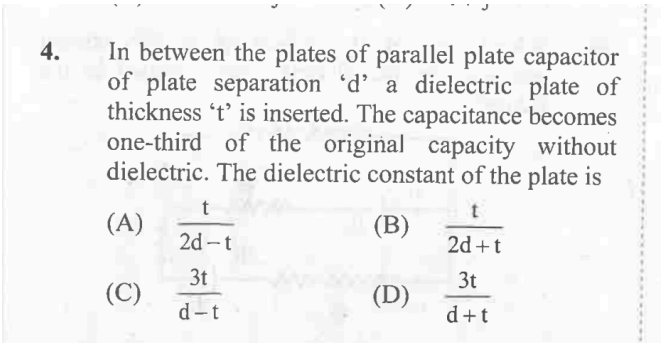
(D) $\hat{i} + \hat{j}$

Phy_q004 key **B** V9_INVALID

Formal answer B gives K = t/(2d+t) < 1, but a dielectric cannot reduce capacitance, so the stated premise is physically impossible.



4. In between the plates of parallel plate capacitor of plate separation 'd' a dielectric plate of thickness 't' is inserted. The capacitance becomes one-third of the original capacity without dielectric. The dielectric constant of the plate is

(A) $\frac{t}{2d-t}$ (B) $\frac{t}{2d+t}$

(C) $\frac{3t}{d-t}$ (D) $\frac{3t}{d+t}$

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(B) $\frac{t}{2d+t}$

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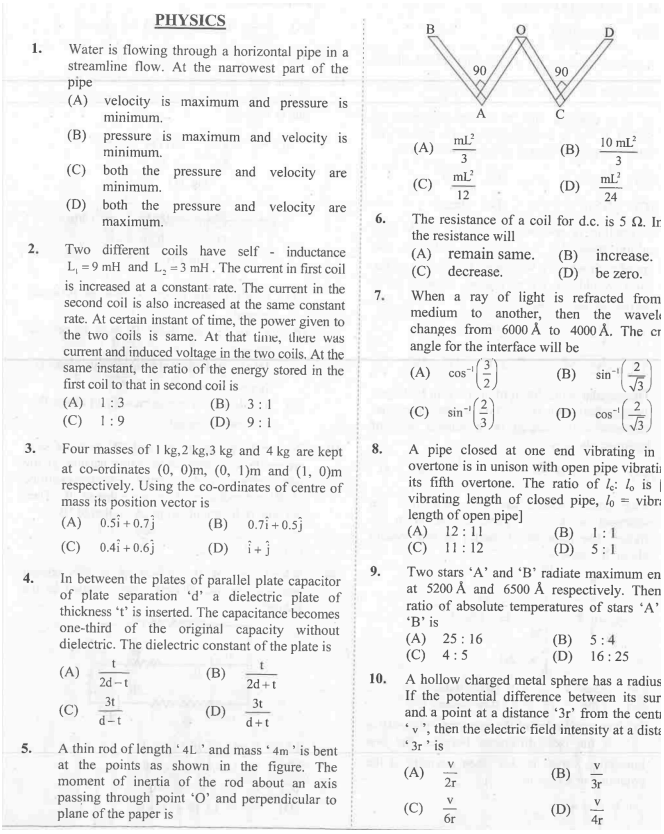
(D) $\frac{3t}{d+t}$

Phy_q005 key **B**



PHYSICS

1. Water is flowing through a horizontal pipe in a streamline flow. At the narrowest part of the pipe
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3. Four masses of 1 kg, 2 kg, 3 kg and 4 kg are kept at co-ordinates (0, 0)m, (0, 1)m and (1, 0)m respectively. Using the co-ordinates of centre of mass its position vector is
(A) $0.5\hat{i}+0.7\hat{j}$ (B) $0.7\hat{i}+0.5\hat{j}$
(C) $0.4\hat{i}+0.6\hat{j}$ (D) $\hat{i}+\hat{j}$

4. In between the plates of parallel plate capacitor of plate separation 'd' a dielectric plate of thickness 't' is inserted. The capacitance becomes one-third of the original capacity without dielectric. The dielectric constant of the plate is
(A) $\frac{t}{2d-t}$ (B) $\frac{t}{2d+t}$
(C) $\frac{3t}{d-t}$ (D) $\frac{3t}{d+t}$

5. A thin rod of length '4L' and mass '4m' is bent at the points as shown in the figure. The moment of inertia of the rod about an axis passing through point 'O' and perpendicular to plane of the paper is

B O D
90 90
A C

(A) $\frac{mL^2}{3}$ (B) $\frac{10\,mL^2}{3}$
(C) $\frac{mL^2}{12}$ (D) $\frac{mL^2}{24}$

6. The resistance of a coil for d.c. is 5 Ω. Ir the resistance will
(A) remain same. (B) increase.
(C) decrease. (D) be zero.

7. When a ray of light is refracted from medium to another, then the wavel changes from 6000 Å to 4000 Å. The cr angle for the interface will be
(A) $\cos^{-1}\left(\frac{3}{2}\right)$ (B) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$
(C) $\sin^{-1}\left(\frac{2}{3}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

8. A pipe closed at one end vibrating in overtone is in unison with open pipe vibrati its fifth overtone. The ratio of l_c: l_o is vibrating length of closed pipe, l_0 = vibr length of open pipe]
(A) 12 : 11 (B) 1 : 1
(C) 11 : 12 (D) 5 : 1

9. Two stars 'A' and 'B' radiate maximum en at 5200 Å and 6500 Å respectively. Then ratio of absolute temperatures of stars 'A' 'B' is
(A) 25 : 16 (B) 5 : 4
(C) 4 : 5 (D) 16 : 25

10. A hollow charged metal sphere has a radius If the potential difference between its sur and a point at a distance '3r' from the centi 'v', then the electric field intensity at a dist '3r' is
(A) $\frac{v}{2r}$ (B) $\frac{v}{3r}$
(C) $\frac{v}{6r}$ (D) $\frac{v}{4r}$

A thin rod of length '$4L$' and mass '$4m$' is bent at the points as shown in the figure. The moment of inertia of the rod about an axis passing through point 'O' and perpendicular to plane of the paper is [figure]

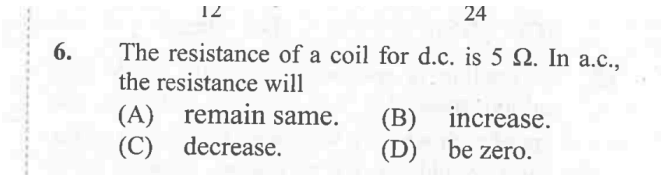
(A) $\frac{mL^2}{3}$

(B) $\frac{10mL^2}{3}$

(C) $\frac{mL^2}{12}$

(D) $\frac{mL^2}{24}$

Phy_q006 key **B**



12 24

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(A) remain same. (B) increase.
(C) decrease. (D) be zero.

The resistance of a coil for d.c. is $5\,\Omega$. In a.c., the resistance will

(A) remain same.

(B) increase.

(C) decrease.

(D) be zero.

Phy_q007 key **A**

(C) decrease. (D) be zero.

7. When a ray of light is refracted from one medium to another, then the wavelength changes from 6000 Å to 4000 Å. The critical angle for the interface will be

(A) $\cos^{-1}\left(\frac{3}{2}\right)$ (B) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(C) $\sin^{-1}\left(\frac{2}{3}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

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(B) $\sin^{-1}\left(\frac{3}{2}\right)$

(C) $\cos^{-1}\left(\frac{2}{3}\right)$

(D) $\cos^{-1}\left(\frac{3}{2}\right)$

Phy_q008 key **C**

8. A pipe closed at one end vibrating in fifth overtone is in unison with open pipe vibrating in its fifth overtone. The ratio of l_c: l_0 is [l_c = vibrating length of closed pipe, l_0 = vibrating length of open pipe]

(A) 12 : 11 (B) 1 : 1

(C) 11 : 12 (D) 5 : 1

A pipe closed at one end vibrating in fifth overtone is in unison with open pipe vibrating in its fifth overtone. The ratio of l_c: l_0 is [l_c = vibrating length of closed pipe, l_0 = vibrating length of open pipe]

(A) $12 : 11$

(B) $1 : 1$

(C) $11 : 12$

(D) $5 : 1$

Phy_q009 key **B**

9. Two stars 'A' and 'B' radiate maximum energy at 5200 Å and 6500 Å respectively. Then the ratio of absolute temperatures of stars 'A' and 'B' is

(A) 25 : 16 (B) 5 : 4

(C) 4 : 5 (D) 16 : 25

Two stars 'A' and 'B' radiate maximum energy at 5200 Å and 6500 Å respectively. Then the ratio of absolute temperatures of stars 'A' and 'B' is

(A) $25 : 16$

(B) $5 : 4$

(C) $4 : 5$

(D) $16 : 25$

Phy_q010 key **C**

(C) 4 : 5 (D) 16 : 25

10. A hollow charged metal sphere has a radius 'r'. If the potential difference between its surface and a point at a distance '3r' from the centre is ' v ', then the electric field intensity at a distance ' 3r ' is

(A) $\frac{v}{2r}$ (B) $\frac{v}{3r}$

(C) $\frac{v}{6r}$ (D) $\frac{v}{4r}$

1

A hollow charged metal sphere has a radius 'r'. If the potential difference between its surface and a point at a distance '$3r$' from the centre is 'v', then the electric field intensity at a distance '$3r$' is

(A) $\frac{v}{2r}$

(B) $\frac{v}{3r}$

(C) $\frac{v}{6r}$

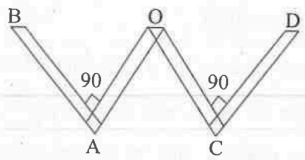
(D) $\frac{v}{4r}$

MHT - CET 2024

nd Biology Total Marks: 200

iemistry

Total Marks: 100



Phy_q011 key **A**

Solved Papers - 2024

11. For a given medium, the speed of light and the polarising angle are ' v ' and ' i_p ' respectively, then (c = speed of light in vaccum)

(A) $v \sin(i_p) = c \cos(i_p)$

(B) $c = v \cot(i_p)$

(C) $v \cos(i_p) = c \sin(i_p)$

(D) $v = c \cos(i_p)$

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(C) $v \cos(i_p) = c \sin(i_p)$

(D) $v = c \cos(i_p)$

Phy_q012 key **C**

12. A particle of mass 'm' is executing S.H.M. about the origin on x-axis with frequency $\sqrt{\frac{ka}{\pi m}}$, where 'k' is a constant and 'a' is the amplitude of S.H.M. If ' x ' is a displacement of a particle, at time 't', potential energy of the particle will be

(A) $\frac{1}{2}kax^2$ (B) πkax^2

(C) $2\pi kax^2$ (D) $2kax^2$

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(A) $\frac{1}{2}kax^2$

(B) πkax^2

(C) $2\pi kax^2$

(D) $2kax^2$

Phy_q013 key **D**

13. A satellite is revolving round the earth with orbital speed 'V_0'. If it stops suddenly, the speed with which it will strike the surface of the earth would be (V_e = escape velocity of a particle on earth's surface)

(A) $\frac{V_e^2}{V_0}$ (B) $2V_0$

(C) $\sqrt{V_e^2 - V_0^2}$ (D) $\sqrt{V_e^2 - 2V_0^2}$

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(B) $2V_0$

(C) $\sqrt{V_e^2 - V_0^2}$

(D) $\sqrt{V_e^2 - 2V_0^2}$

Phy_q014 key **B**

14. The angular momentum of electron in hydrogen atom in first orbit is 'L'. The change in angular momentum if electron is in second orbit of hydrogen atom is

(A) 2L (B) L (C) $\frac{L}{2}$ (D) 4L

The angular momentum of electron in hydrogen atom in first orbit is 'L'. The change in angular momentum if electron is in second orbit of hydrogen atom is

(A) $2L$

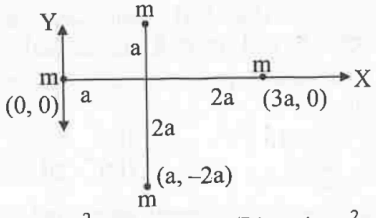
(B) L

(C) $\frac{L}{2}$

(D) $4L$

Phy_q015 key **C**

15. Four point masses, each of mass 'm' are arranged in X - Y plane as shown in the figure. The moment of inertia of this system about X - axis is



(A) 3 ma^2 (B) 4 ma^2

(C) 5 ma^2 (D) 6 ma^2

16. In biprism experiment the maximum intensity is 'I_0'. If the path difference between the two interfering waves is 'λ/4' then intensity at the point on the screen is

$\left[\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}\right]$

Four point masses, each of mass 'm' are arranged in X - Y plane as shown in the figure. The moment of inertia of this system about X - axis is [figure]

(A) $3ma^2$

(B) $4ma^2$

(C) $5ma^2$

(D) $6ma^2$

l the
vely,

(A) $\frac{I_0}{4}$ (B) $\frac{I_0}{3}$

(C) $\frac{I_0}{2}$ (D) I_0

Phy_q016 key **C**

Solved Papers - 2024

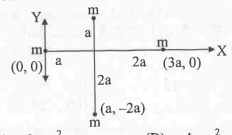
11. For a given medium, the speed of light and the polarising angle are ' v ' and ' i_p ' respectively, then (c = speed of light in vaccum)
(A) $v \sin(i_p) = c \cos(i_p)$
(B) $c = v \cot(i_p)$
(C) $v \cos(i_p) = c \sin(i_p)$
(D) $v = c \cos(i_p)$

12. A particle of mass 'm' is executing S.H.M. about the origin on x-axis with frequency $\sqrt{\frac{ka}{\pi m}}$, where 'k' is a constant and 'a' is the amplitude of S.H.M. If ' x ' is a displacement of a particle, at time 't', potential energy of the particle will be
(A) $\frac{1}{2}kax^2$ (B) πkax^2
(C) $2\pi kax^2$ (D) $2kax^2$

13. A satellite is revolving round the earth with orbital speed ' V_0 '. If it stops suddenly, the speed with which it will strike the surface of the earth would be (V_e = escape velocity of a particle on earth's surface)
(A) $\frac{V_e^2}{V_0}$ (B) $2V_0$
(C) $\sqrt{V_e^2 - V_0^2}$ (D) $\sqrt{V_e^2 - 2V_0^2}$

14. The angular momentum of electron in hydrogen atom in first orbit is 'L'. The change in angular momentum if electron is in second orbit of hydrogen atom is
(A) 2L (B) L (C) $\frac{L}{2}$ (D) 4L

15. Four point masses, each of mass ' m ' are arranged in X - Y plane as shown in the figure. The moment of inertia of this system about X - axis is



(A) 3 ma^2 (B) 4 ma^2
(C) 5 ma^2 (D) 6 ma^2

16. In biprism experiment the maximum intensity is ' I_0 '. If the path difference between the two interfering waves is 'λ/4' then intensity at the point on the screen is
$\left[\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}\right]$
(A) $\frac{I_0}{4}$ (B) $\frac{I_0}{3}$
(C) $\frac{I_0}{2}$ (D) I_0

17. Work done to get ' n ' spherical drop size from a single spherical drop o proportional to
(A) $\left(\frac{1}{n^{2/3}} - 1\right)$ (B) $\left(\frac{1}{n^{1/3}}\right.$
(C) $n^{1/3} - 1$ (D) $n^{4/3}$ –

18. Diode and resistance are connected a figure. Out of the following statem one is TRUE?

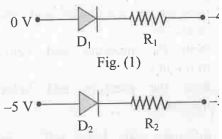


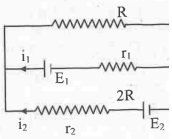
Fig. (1)

Fig. (2)

(A) Diode D_1 and diode D_2 both a biased.
(B) Diode D_1 . and diode D_2 both biased.
(C) Diode D_1 is forward biased ar is reverse biased.
(D) Diode D_1 is reverse biased an is forward biased.

19. Two uniform strings 'A' and 'B' m are made to vibrate under same te first overtone of 'A' is equal to sec of 'B' and radius of 'A' is twice that the ratio of length of string 'A' to tha
(A) 2:1 (B) 3:4
(C) 3:2 (D) 1:3

20. Which one of the following is equation for the electric circuit s figure?



(A) $E_1 - (i_1 + i_2)R + i_1 r_1 = 0$
(B) $E_1 - (i_1 + i_2)R - i_1 r_1 = 0$
(C) $E_2 - i_2 r_2 - E_1 - i_1 r_1 = 0$
(D) $E_2 - (i_1 + i_2)R + i_2 r_2 = 0$

In biprism experiment the maximum intensity is 'I_0'. If the path difference between the two interfering waves is '$\lambda/4$' then intensity at the point on the screen is $\left[\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}\right]$

(A) $\frac{I_0}{4}$
(B) $\frac{I_0}{3}$
(C) $\frac{I_0}{2}$
(D) I_0

Phy_q017 key **C**

2

17. Work done to get ' n ' spherical drops of equal size from a single spherical drop of water, is proportional to
(A) $\left(\frac{1}{n^{2/3}} - 1\right)$ (B) $\left(\frac{1}{n^{1/3}} - 1\right)$
(C) $n^{1/3} - 1$ (D) $n^{4/3} - 1$

Work done to get 'n' spherical drops of equal size from a single spherical drop of water, is proportional to

(A) $\left(\frac{1}{n^{2/3}} - 1\right)$
(B) $\left(\frac{1}{n^{1/3}} - 1\right)$
(C) $n^{1/3} - 1$
(D) $n^{4/3} - 1$

Phy_q018 key **C**

18. Diode and resistance are connected as shown in figure. Out of the following statements which one is TRUE?

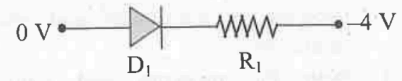


Fig. (1)

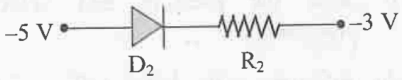


Fig. (2)

(A) Diode D_1 and diode D_2 both are forward biased.
(B) Diode D_1 and diode D_2 both are reverse biased.
(C) Diode D_1 is forward biased and diode D_2 is reverse biased.
(D) Diode D_1 is reverse biased and diode D_2 is forward biased.

Diode and resistance are connected as shown in figure. Out of the following statements which one is TRUE? [figure]

(A) Diode D_1 and diode D_2 both are forward biased.

(B) Diode D_1 and diode D_2 both are reverse biased.

(C) Diode D_1 is forward biased and diode D_2 is reverse biased.

(D) Diode D_1 is reverse biased and diode D_2 is forward biased.

Phy_q019 key **D**

is forward biased.

19. Two uniform strings 'A' and 'B' made of steel are made to vibrate under same tension. If the first overtone of 'A' is equal to second overtone of 'B' and radius of 'A' is twice that of 'B'. Then the ratio of length of string 'A' to that of 'B' is

(A) 2:1 (B) 3:4
(C) 3:2 (D) 1:3

Two uniform strings 'A' and 'B' made of steel are made to vibrate under same tension. If the first overtone of 'A' is equal to second overtone of 'B' and radius of 'A' is twice that of 'B'. Then the ratio of length of string 'A' to that of 'B' is

(A) $2:1$

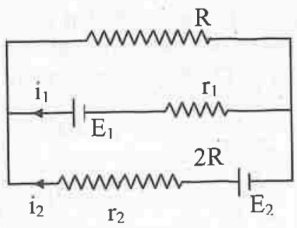
(B) $3:4$

(C) $3:2$

(D) $1:3$

Phy_q020 key **B**

20. Which one of the following is the correct equation for the electric circuit shown in the figure?



(A) $E_1 - (i_1 + i_2)R + i_1 r_1 = 0$
(B) $E_1 - (i_1 + i_2)R - i_1 r_1 = 0$
(C) $E_2 - i_2 r_2 - E_1 - i_1 r_1 = 0$
(D) $E_2 - (i_1 + i_2)R + i_2 r_2 = 0$

Which one of the following is the correct equation for the electric circuit shown in the figure? [figure]
(A) $E_1 - (i_1 + i_2)R + i_1 r_1 = 0$
(B) $E_1 - (i_1 + i_2)R - i_1 r_1 = 0$
(C) $E_2 - i_2 r_2 - E_1 - i_1 r_1 = 0$
(D) $E_2 - (i_1 + i_2)R + i_2 r_2 = 0$

Phy_q021 key **B**

21. A pendulum is performing simple harmonic motion. The acceleration of the bob is 20 cm s^{-2} at a distance of 5 cm from mean position. The time period of oscillation is
(A) 2 s (B) π s
(C) 2π s (D) 1 s

A pendulum is performing simple harmonic motion. The acceleration of the bob is $20\ \text{cm}\,\text{s}^{-2}$ at a distance of 5 cm from mean position. The time period of oscillation is
(A) 2 s
(B) π s
(C) 2π s
(D) 1 s

Phy_q022 key **C**

22. A spherical body of radius 'r' radiates power 'P' at T kelvin. If the radius s halved and the temperature doubled the power radiated in the same time 't' will be
(A) $\frac{P}{2}$ (B) 2P
(C) 4P (D) 8P

A spherical body of radius 'r' radiates power 'P' at T kelvin. If the radius s halved and the temperature doubled the power radiated in the same time 't' will be
(A) $\frac{P}{2}$
(B) $2P$
(C) $4P$
(D) $8P$

Phy_q023 key **C**

23. For a common emitter transistor configuration the ratio of $\frac{I_C}{I_E} = 0.96$, then the current gain in this configuration is
(A) 6 (B) 12
(C) 24 (D) 48

For a common emitter transistor configuration the ratio of $\frac{I_C}{I_E} = 0.96$, then the current gain in this configuration is
(A) 6
(B) 12
(C) 24
(D) 48

Phy_q024 key **B**

24. A stone is thrown upward with a speed 'u' from the top of a tower reaches the ground with velocity '3u'. The height of the tower is (g = acceleration due to gravity)

(A) $\frac{3u^2}{g}$ (B) $\frac{4u^2}{g}$

(C) $\frac{6u^2}{g}$ (D) $\frac{9u^2}{g}$

A stone is thrown upward with a speed 'u' from the top of a tower reaches the ground with velocity '$3u$'. The height of the tower is (g = acceleration due to gravity)

(A) $\frac{3u^2}{g}$

(B) $\frac{4u^2}{g}$

(C) $\frac{6u^2}{g}$

(D) $\frac{9u^2}{g}$

Phy_q025 key **B**

25. A metal surface is illuminated by light of two different wavelengths 207 nm and 414 nm. The maximum speeds of photoelectrons corresponding to these wavelengths are u_1 and u_2 respectively with $u_1 : u_2 = 2:1$. The work function of the metal is $(hc = 1242\ \text{eV nm})$

(A) 1.6 eV (B) 2.0 eV

(C) 2.4 eV (D) 3.0 eV

A metal surface is illuminated by light of two different wavelengths $207\ \text{nm}$ and $414\ \text{nm}$. The maximum speeds of photoelectrons corresponding to these wavelengths are u_1 and u_2 respectively with $u_1 : u_2 = 2 : 1$. The work function of the metal is $(hc = 1242\ \text{eV nm})$

(A) $1.6\ \text{eV}$

(B) $2.0\ \text{eV}$

(C) $2.4\ \text{eV}$

(D) $3.0\ \text{eV}$

Phy_q026 key **B**

26. In thermodynamic process, which of the following statements is not true?

(A) In an adiabatic process, the system is insulated from the surroundings.

(B) In an isochoric process, the pressure remains constant.

(C) In an isothermal process, the temperature remains constant.

(D) In an adiabatic process, pv^γ = constant.

In thermodynamic process, which of the following statements is not true?

(A) In an adiabatic process, the system is insulated from the surroundings.

(B) In an isochoric process, the pressure remains constant.

(C) In an isothermal process, the temperature remains constant.

(D) In an adiabatic process, $pv^\gamma = \text{constant}$.

Phy_q027 key **C**

22 April (Shift – I)

27. A charged particle is subjected to acceleration in a cyclotron which consists of two dees 'D_1' and 'D_2'. The charged particle undergoes increase in its speed.

(A) inside D_1, D_2 and the gap between dees.

(B) only inside D_1.

(C) only in the gap between D_1 and D_2.

(D) only inside D_2.

A charged particle is subjected to acceleration in a cyclotron which consists of two dees 'D_1' and 'D_2'. The charged particle undergoes increase in its speed.

(A) inside D_1, D_2 and the gap between dees.

(B) only inside D_1.

(C) only in the gap between D_1 and D_2.

(D) only inside D_2.

Phy_q028 key **C**

28. In case of well of death which is a vertical cylindrical wall of radius 'r' inside which vehicle is driven in horizontal circles. If ' m ' is mass of vehicle, ' V ' is the velocity and ' μ_s ' is the coefficient of static friction between the wheels of vehicle and walls then correct relation is [g = acceleration due to gravity]

(A) $V^2 \le \frac{rg}{\mu_s}$ (B) $V \le \frac{rg}{\mu_s}$

(C) $V^2 \ge \frac{rg}{\mu_s}$ (D) $V \ge \frac{rg}{\mu_s}$

In case of well of death which is a vertical cylindrical wall of radius 'r' inside which vehicle is driven in horizontal circles. If 'm' is mass of vehicle, 'V' is the velocity and 'μ_s' is the coefficient of static friction between the wheels of vehicle and walls then correct relation is [g = acceleration due to gravity]

(A) $V^2 \le \frac{rg}{\mu_s}$

(B) $V \le \frac{rg}{\mu_s}$

(C) $V^2 \ge \frac{rg}{\mu_s}$

(D) $V \ge \frac{rg}{\mu_s}$

Phy_q029 key **C**

29. In an oscillating LC circuit, the maximum charge on the capacitor is ' Q '. When the energy is stored equally between the electric and magnetic fields, the instantaneous charge on the capacitor ' q ' is

(A) Q (B) $\frac{Q}{2}$

(C) $\frac{Q}{\sqrt{2}}$ (D) $\frac{Q}{\sqrt{3}}$

In an oscillating LC circuit, the maximum charge on the capacitor is 'Q'. When the energy is stored equally between the electric and magnetic fields, the instantaneous charge on the capacitor 'q' is

(A) Q

(B) $\frac{Q}{2}$

(C) $\frac{Q}{\sqrt{2}}$

(D) $\frac{Q}{\sqrt{3}}$

Phy_q030 key **D**

30. Two particles execute S.H.M. of same amplitude and frequency along the same straight line path. They pass each other when going in opposite directions, each time their displacement is half the amplitude. The phase difference between them is $(\sin 30^\circ = 0 \cdot 5)$

(A) $\frac{\pi}{6}$ (B) $\frac{5\pi}{6}$

(C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

Two particles execute S.H.M. of same amplitude and frequency along the same straight line path. They pass each other when going in opposite directions, each time their displacement is half the amplitude. The phase difference between them is $(\sin 30^\circ = 0.5)$

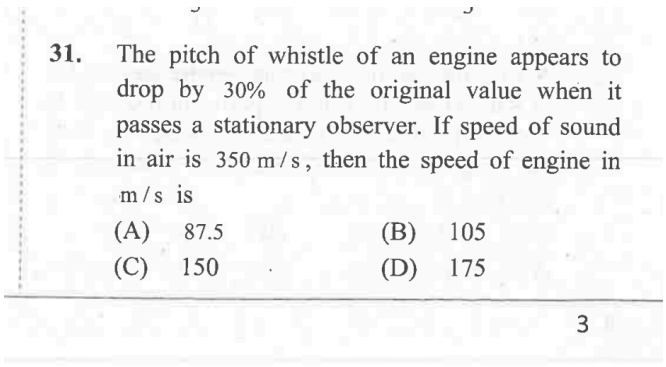
(A) $\frac{\pi}{6}$

(B) $\frac{5\pi}{6}$

(C) $\frac{\pi}{3}$

(D) $\frac{2\pi}{3}$

Phy_q031 key **C**



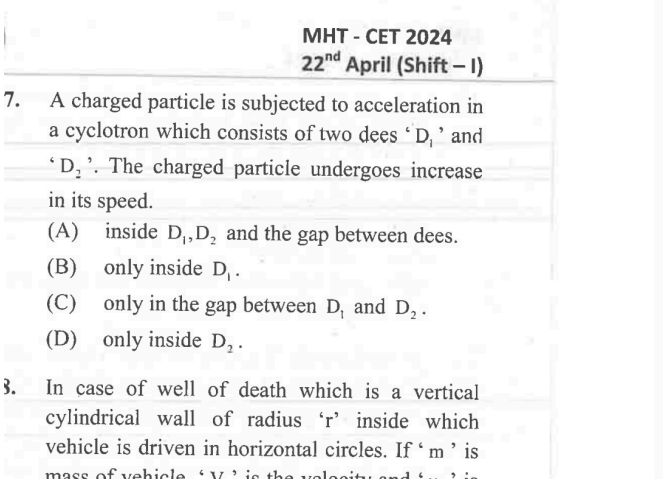
31. The pitch of whistle of an engine appears to drop by 30% of the original value when it passes a stationary observer. If speed of sound in air is 350 m/s, then the speed of engine in m/s is

(A) 87.5 (B) 105
(C) 150 (D) 175

3

The pitch of whistle of an engine appears to drop by 30% of the original value when it passes a stationary observer. If speed of sound in air is 350 m / s, then the speed of engine in m / s is

(A) 87.5
(B) 105
(C) 150
(D) 175



MHT - CET 2024
22nd April (Shift – I)

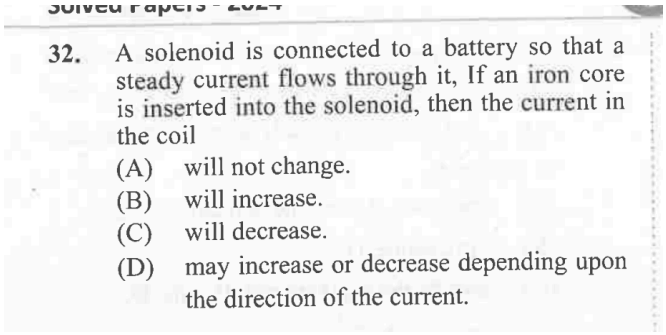
7. A charged particle is subjected to acceleration in a cyclotron which consists of two dees 'D_1' and 'D_2'. The charged particle undergoes increase in its speed.

(A) inside D_1, D_2 and the gap between dees.
(B) only inside D_1.
(C) only in the gap between D_1 and D_2.
(D) only inside D_2.

8. In case of well of death which is a vertical cylindrical wall of radius 'r' inside which vehicle is driven in horizontal circles. If 'm' is mass of vehicle, 'V' is the velocity and '...' is

Phy_q032 key **C** V9_NEEDS_HUMAN

Steady DC current is fixed by V/R, so the core should not change it (A); key C is doubtful, but confidence is below 0.8 and the key may assume a transient.



32. A solenoid is connected to a battery so that a steady current flows through it, If an iron core is inserted into the solenoid, then the current in the coil

(A) will not change.
(B) will increase.
(C) will decrease.
(D) may increase or decrease depending upon the direction of the current.

A solenoid is connected to a battery so that a steady current flows through it, If an iron core is inserted into the solenoid, then the current in the coil

(A) will not change.
(B) will increase.
(C) will decrease.
(D) may increase or decrease depending upon the direction of the current.

Phy_q033 key **D**

33. A monoatomic ideal gas, initially at temperature 'T_1' is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature 'T_2' by releasing the piston suddenly. L_1 and L_2 are the lengths of the gas columns before and after the expansion respectively. Then $\frac{T_1}{T_2}$ is

(A) $\sqrt{\frac{L_1}{L_2}}$ (B) $\sqrt{\frac{L_2}{L_1}}$

(C) $\left(\frac{L_1}{L_2}\right)^{2/3}$ (D) $\left(\frac{L_2}{L_1}\right)^{2/3}$

A monoatomic ideal gas, initially at temperature 'T_1' is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature 'T_2' by releasing the piston suddenly. L_1 and L_2 are the lengths of the gas columns before and after the expansion respectively. Then $\frac{T_1}{T_2}$ is

(A) $\sqrt{\frac{L_1}{L_2}}$

(B) $\sqrt{\frac{L_2}{L_1}}$

(C) $\left(\frac{L_1}{L_2}\right)^{2/3}$

(D) $\left(\frac{L_2}{L_1}\right)^{2/3}$

Phy_q034 key **A**

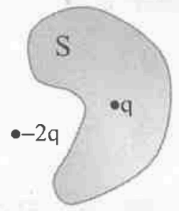
34. In the diagram, the total electric flux through the closed surface 'S' is
[Given q = charge
ε_0 = permittivity of free space]

(A) $\frac{q}{\varepsilon_0}$

(B) $\frac{-2q}{\varepsilon_0}$

(C) $\frac{-q}{\varepsilon_0}$

(D) $\frac{3q}{\varepsilon_0}$



In the diagram, the total electric flux through the closed surface 'S' is [figure] [Given q = charge ϵ_0 = permittivity of free space]

(A) $\frac{q}{\epsilon_0}$

(B) $\frac{-2q}{\epsilon_0}$

(C) $\frac{-q}{\epsilon_0}$

(D) $\frac{3q}{\epsilon_0}$

Phy_q035 key **C**

35. If an electron in hydrogen atom jumps from 3^{rd} orbit to 2^{nd} orbit it emits a photon of wavelength 'λ'. When it emits a photon from 4^{th} orbit to 3^{rd} orbit then the corresponding wavelength of emitted photon will be

(A) $\frac{16}{25}\lambda$ (B) $\frac{9}{16}\lambda$

(C) $\frac{20}{7}\lambda$ (D) $\frac{20}{13}\lambda$

If an electron in hydrogen atom jumps from 3^{rd} orbit to 2^{nd} orbit it emits a photon of wavelength λ. When it emits a photon from 4^{th} orbit to 3^{rd} orbit then the corresponding wavelength of emitted photon will be

(A) $\frac{16}{25}\lambda$

(B) $\frac{9}{16}\lambda$

(C) $\frac{20}{7}\lambda$

(D) $\frac{20}{13}\lambda$

Phy_q036 key **C**

36. A solenoid having 400 turns/metre has a core of a material with relative permeability 300. If a current of 0.5 A is passed through it, then the magnetisation of the core material is nearly

(A) 6×10^{2} A/m (B) 6×10^{3} A/m

(C) 6×10^{4} A/m (D) 6×10^{5} A/m

4

A solenoid having 400 turns/metre has a core of a material with relative permeability 300. If a current of $0.5\ \mathrm{A}$ is passed through it, then the magnetisation of the core material is nearly

(A) $6 \times 10^{2}\ \mathrm{A/m}$

(B) $6 \times 10^{3}\ \mathrm{A/m}$

(C) $6 \times 10^{4}\ \mathrm{A/m}$

(D) $6 \times 10^{5}\ \mathrm{A/m}$

Phy_q037 key **D**

37. The volume of a block of metal at 30°C changes by 0.12% when its temperature is increased to 70°C. The coefficient of linear expansion of the metal is

(A) 2×10^{-5} °C (B) 3×10^{-5} °C

(C) 4×10^{-5} °C (D) 1×10^{-5} °C

The volume of a block of metal at $30^{\circ}\mathrm{C}$ changes by 0.12% when its temperature is increased to $70^{\circ}\mathrm{C}$. The coefficient of linear expansion of the metal is

(A) $2 \times 10^{-5}\ ^{\circ}\mathrm{C}^{-1}$

(B) $3 \times 10^{-5}\ ^{\circ}\mathrm{C}^{-1}$

(C) $4 \times 10^{-5}\ ^{\circ}\mathrm{C}^{-1}$

(D) $1 \times 10^{-5}\ ^{\circ}\mathrm{C}^{-1}$

Phy_q038 key **C**

38. A galvanometer of resistance 'G' can be converted into a voltmeter of range (0 – V) volt by connecting a resistance 'R' in series with it. The resistance 'R required to change its range from $\left(0-\frac{V}{4}\right)$ volt will be

(A) $\frac{R-G}{2}$ (B) $\frac{R-2G}{3}$

(C) $\frac{R-3G}{4}$ (D) $\frac{4R-3G}{5}$

A galvanometer of resistance 'G' can be converted into a voltmeter of range $(0 - V)$ volt by connecting a resistance 'R' in series with it. The resistance 'R' required to change its range from $\left(0 - \frac{V}{4}\right)$ volt will be

(A) $\frac{R-G}{2}$

(B) $\frac{R-2G}{3}$

(C) $\frac{R-3G}{4}$

(D) $\frac{4R-3G}{5}$

Phy_q039 key **C**

39. A light metal disc of radius 'r' floats on water surface and bends the surface downwards along the perimeter making an angle 'θ' with the vertical edge of the disc. If the weight of water displaced by the disc is 'W', the weight of the metal disc is [T = surface tension of water]

(A) $W - 2\pi rT\cos\theta$ (B) $2\pi rT + W$

(C) $2\pi rT\cos\theta + W$ (D) $2\pi rT\cos\theta - W$

A light metal disc of radius 'r' floats on water surface and bends the surface downwards along the perimeter making an angle 'θ' with the vertical edge of the disc. If the weight of water displaced by the disc is 'W', the weight of the metal disc is [T = surface tension of water]

(A) $W - 2\pi rT\cos\theta$

(B) $2\pi rT + W$

(C) $2\pi rT\cos\theta + W$

(D) $2\pi rT\cos\theta - W$

Phy_q040 key **A**

40. The magnetic field at the centre of a current carrying circular coil of area 'A' is 'B'. The magnetic moment of the coil is (μ_0 = permeability of free space)

(A) $\frac{2B}{\mu_0}\sqrt{\frac{A^3}{\pi}}$ (B) $\frac{BA^2}{4\mu_0\pi}$

(C) $\frac{2\pi}{\mu_0}\sqrt{A^3}$ (D) $\frac{\mu_0}{2B}\sqrt{\frac{A^3}{\pi}}$

The magnetic field at the centre of a current carrying circular coil of area 'A' is 'B'. The magnetic moment of the coil is (μ_0 = permeability of free space)

(A) $\frac{2B}{\mu_0}\sqrt{\frac{A^3}{\pi}}$

(B) $\frac{BA^2}{4\mu_0\pi}$

(C) $\frac{2\pi}{\mu_0}\sqrt{A^3}$

(D) $\frac{\mu_0}{2B}\sqrt{\frac{A^3}{\pi}}$

Phy_q041 key **D**

41. A parallel combination of two capacitors of capacities 'C' and '$\frac{C}{3}$' respectively is connected across a battery of 12 volt. When both capacitors are fully charged, the charge and energy stored in them is Q_1, Q_2 and E_1, E_2 respectively. Then the ratio of $(E_1 - E_2)$ to $(Q_1 - Q_2)$ is

(A) 1 : 8 (B) 1 : 6

(C) 8 : 1 (D) 6 : 1

A parallel combination of two capacitors of capacities 'C' and $\frac{C}{3}$, respectively is connected across a battery of 12 volt. When both capacitors are fully charged, the charge and energy stored in them is Q_1, Q_2 and E_1, E_2 respectively. Then the ratio of $(E_1 - E_2)$ to $(Q_1 - Q_2)$ is

(A) $1 : 8$

(B) $1 : 6$

(C) $8 : 1$

(D) $6 : 1$

Phy_q042 key **C**

42. A transparent sphere of refractive index 'μ' and radius of curvature 'R' is kept in air. A point object is placed at a distance 'd' from the surface of the sphere so that the real image is formed at the same distance 'd' from exactly opposite side of the sphere. The distance 'd' is

(A) $\frac{\mu}{R}$ (B) $R(\mu - 1)$

(C) $\frac{R}{(\mu - 1)}$ (D) $\frac{R}{(\mu + 1)}$

A transparent sphere of refractive index μ and radius of curvature 'R' is kept in air. A point object is placed at a distance 'd' from the surface of the sphere so that the real image is formed at the same distance 'd' from exactly opposite side of the sphere. The distance 'd' is

(A) $\frac{\mu}{R}$

(B) $R(\mu - 1)$

(C) $\frac{R}{(\mu - 1)}$

(D) $\frac{R}{(\mu + 1)}$

Phy_q043 key **D**

43. The instantaneous value of current in an a.c. circuit is $I = 2\sin\left[100\pi t + \frac{\pi}{3}\right]$ A. The current will be maximum for the first time at $(\sin 90^0 = 1)$

(A) $\frac{1}{100}$ s (B) $\frac{1}{200}$ s

(C) $\frac{1}{400}$ s (D) $\frac{1}{600}$ s

The instantaneous value of current in an a.c. circuit is $I = 2\sin\left[100\pi t + \frac{\pi}{3}\right]$ A. The current will be maximum for the first time at $(\sin 90^\circ = 1)$

(A) $\frac{1}{100}$ s

(B) $\frac{1}{200}$ s

(C) $\frac{1}{400}$ s

(D) $\frac{1}{600}$ s

Phy_q044 key **D**

44. A fixed mass of gas at constant pressure occupies a volume 'V'. The gas undergoes a rise in temperature so that the r.m.s. velocity of the molecule is doubled. The new volume will be

(A) $\frac{V}{2}$ (B) $\frac{V}{\sqrt{2}}$

(C) 2 V (D) 4V

A fixed mass of gas at constant pressure occupies a volume 'V'. The gas undergoes a rise in temperature so that the r.m.s. velocity of the molecule is doubled. The new volume will be

(A) $\frac{V}{2}$

(B) $\frac{V}{\sqrt{2}}$

(C) $2V$

(D) $4V$

Phy_q045 key **A**

45. The Boolean expression for X – OR gate $C = (A \oplus B)$ is equivalent to

(A) $(\bar{A}\cdot B)+(A\cdot\bar{B})$ (B) $A+(\bar{A}\cdot\bar{B})$

(C) $(A\cdot B)+\bar{B}$ (D) $(A\cdot B)+(\bar{A}\cdot\bar{B})$

The Boolean expression for X-OR gate $C = (A \oplus B)$ is equivalent to

(A) $(\overline{A}\cdot B)+(A\cdot\overline{B})$

(B) $A+(\overline{A}\cdot\overline{B})$

(C) $(A\cdot B)+\overline{B}$

(D) $(A\cdot B)+(\overline{A}\cdot\overline{B})$

Phy_q046 key **B**

46. The equation of simple harmonic progressive wave is given by $y = A\sin(100\pi t - 4x)$. The distance between two particles having a phase difference of $\left(\frac{\pi}{4}\right)^c$ is

(A) $\frac{\pi}{18}$ m (B) $\frac{\pi}{16}$ m

(C) $\frac{\pi}{9}$ m (D) $\frac{\pi}{3}$ m

The equation of simple harmonic progressive wave is given by $y = A\sin(100\pi t - 4x)$. The distance between two particles having a phase difference of $\left(\frac{\pi}{4}\right)^c$ is

(A) $\frac{\pi}{18}$ m

(B) $\frac{\pi}{16}$ m

(C) $\frac{\pi}{9}$ m

(D) $\frac{\pi}{3}$ m

Phy_q047 key **B**

47. A photon and an electron have an equal energy 'E'. The ratio of wavelength 'λ_P' of photon to that of electron 'λ_e' is proportional to

(A) $\sqrt{E}$ (B) $\frac{1}{\sqrt{E}}$

(C) $\frac{1}{E}$ (D) E

A photon and an electron have an equal energy 'E'. The ratio of wavelength λ_p of photon to that of electron λ_e is proportional to

(A) $\sqrt{E}$

(B) $\frac{1}{\sqrt{E}}$

(C) $\frac{1}{E}$

(D) E

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8. The height at which the acceleration due to gravity becomes $\frac{g}{4}$ in terms of R is [R = the radius of the earth]

(A) $\frac{R}{\sqrt{2}}$ (B) R

(C) $\sqrt{2}$ R (D) 2R

9. In a biprism experiment, fifth dark fringe is obtained at a point. A thin transparent film of refractive index 'μ' is placed in one of the interfering paths. Now 7th bright fringe is obtained at the same point. If 'λ' is the wavelength of light used, the thickness of film is equal to

Phy_q048 key **B**

22nd April (Shift – I)

48. The height at which the acceleration due to gravity becomes $\frac{g}{4}$ in terms of R is [R = the radius of the earth]

(A) $\frac{R}{\sqrt{2}}$ (B) R

(C) $\sqrt{2}$ R (D) 2R

The height at which the acceleration due to gravity becomes $\frac{g}{4}$ in terms of R is [R = the radius of the earth]

(A) $\frac{R}{\sqrt{2}}$

(B) R

(C) $\sqrt{2}R$

(D) $2R$

Phy_q049 key **D**

49. In a biprism experiment, fifth dark fringe is obtained at a point. A thin transparent film of refractive index 'μ' is placed in one of the interfering paths. Now 7th bright fringe is obtained at the same point. If 'λ' is the wavelength of light used, the thickness of film is equal to

(A) $1.5(\mu-1)\lambda$ (B) $\frac{1.5\lambda}{(\mu-1)}$

(C) $2.5(\mu-1)\lambda$ (D) $\frac{2.5\lambda}{(\mu-1)}$

In a biprism experiment, fifth dark fringe is obtained at a point. A thin transparent film of refractive index 'μ' is placed in one of the interfering paths. Now 7th bright fringe is obtained at the same point. If 'λ' is the wavelength of light used, the thickness of film is equal to

(A) $1.5(\mu-1)\lambda$

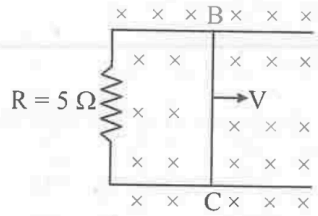
(B) $\frac{1.5\lambda}{(\mu-1)}$

(C) $2.5(\mu-1)\lambda$

(D) $\frac{2.5\lambda}{(\mu-1)}$

Phy_q050 key **D**

50. Figure shows a rectangular frame situated in a constant magnetic field. A wire BC of length 1 m is moved out with velocity 4 m/s. Magnetic field strength is 0.15 T. Force acting on the wire BC is



(A) 18 N (B) 1.8 N
(C) 0.18 N (D) 0.018 N

Figure shows a rectangular frame situated in a constant magnetic field. A wire BC of length $1\ \mathrm{m}$ is moved out with velocity $4\ \mathrm{m}/\mathrm{s}$. Magnetic field strength is $0.15\ \mathrm{T}$. Force acting on the wire BC is [figure]

(A) $18\ \mathrm{N}$

(B) $1.8\ \mathrm{N}$

(C) $0.18\ \mathrm{N}$

(D) $0.018\ \mathrm{N}$

Che_q001 key **C**

1. Which element from following does NOT exhibit magnetic moment in +1 state?

(A) Zn (B) Co
(C) Cu (D) Mn

Which element from following does NOT exhibit magnetic moment in +1 state?

(A) Zn

(B) Co

(C) Cu

(D) Mn

Che_q002 key **B**

2. The entropy of vaporisation of benzene is 85 J K^{-1} mol^{-1}. When 117 g of benzene vaporises at its boiling point, what is entropy change of surrounding if process is at equilibrium?

(A) –85 J K^{-1}
(B) –85 × 1.5 J K^{-1}
(C) 85 × 1.5 J K^{-1}
(D) 42.5 J K^{-1}

The entropy of vaporisation of benzene is $85\ \mathrm{J\,K^{-1}\,mol^{-1}}$. When 117 g of benzene vaporises at its boiling point, what is entropy change of surrounding if process is at equilibrium?

(A) $-85\ \mathrm{J\,K^{-1}}$

(B) $-85 \times 1.5\ \mathrm{J\,K^{-1}}$

(C) $85 \times 1.5\ \mathrm{J\,K^{-1}}$

(D) $42.5\ \mathrm{J\,K^{-1}}$

Che_q003 key **A**

Solved Papers - 2024

3. Which from the following equations represents the relation between solubility (mol L^{-1}) and solubility product for the salt B_2A?

(A) $S=\left(\frac{K_{sp}}{4}\right)^{1/3}$ (B) $S=(4\,K_{sp})^{1/3}$
(C) $S=\left(\frac{K_{sp}}{3}\right)^{1/3}$ (D) $S=(3\,K_{sp})^{1/3}$

Which from the following equations represents the relation between solubility (mol $\mathrm{L^{-1}}$) and solubility product for the salt $\mathrm{B_2\,A}$?

(A) $S=\left(\frac{K_{sp}}{4}\right)^{1/3}$

(B) $S=(4\,K_{sp})^{1/3}$

(C) $S=\left(\frac{K_{sp}}{3}\right)^{1/3}$

(D) $S=(3\,K_{sp})^{1/3}$

Che_q004 key **C**

4. What is the number of chiral carbon atoms and number of formyl groups respectively present in ribose?
(A) 2 and 2 (B) 2 and 1
(C) 3 and 1 (D) 3 and 2

What is the number of chiral carbon atoms and number of formyl groups respectively present in ribose?
(A) 2 and 2
(B) 2 and 1
(C) 3 and 1
(D) 3 and 2

Che_q005 key **A**

5. Calculate the Henry's law constant at 25 °C if solubility of gas in liquid is 2.1×10^{-2} mol dm^{-3} at 0.18 bar.
(A) 0.1166 mol dm^{-3} bar^{-1}
(B) 0.1445 mol dm^{-3} bar^{-1}
(C) 0.1730 mol dm^{-3} bar^{-1}
(D) 0.2014 mol dm^{-3} bar^{-1}

Calculate the Henry's law constant at $25^\circ C$ if solubility of gas in liquid is $2.1 \times 10^{-2} \text{mol dm}^{-3}$ at 0.18 bar.
(A) $0.1166\ \text{mol dm}^{-3}\text{bar}^{-1}$
(B) $0.1445\ \text{mol dm}^{-3}\text{bar}^{-1}$
(C) $0.1730\ \text{mol dm}^{-3}\text{bar}^{-1}$
(D) $0.2014\ \text{mol dm}^{-3}\text{bar}^{-1}$

Che_q006 key **B**

6. What is the name of isopropyl alcohol according to carbinol system?
(A) Methyl carbinol
(B) Dimethyl carbinol
(C) Ethyl carbinol
(D) Propyl carbinol

What is the name of isopropyl alcohol according to carbinol system?
(A) Methyl carbinol
(B) Dimethyl carbinol
(C) Ethyl carbinol
(D) Propyl carbinol

Che_q007 key **B** OPTION_EMPTY

7. What is the frequency of red light having wave length 750 nm?
(A) 3.0×10^{14} Hz (B) 4.0×10^{14} Hz
(C) 7.5×10^{14} Hz (D) 9.0×10^{14} Hz

What is the frequency of red light having wave length 750 nm?
(A) 3.0×10^{14} Hz
(B) 4.0×10^{14} Hz
(C) 7.5×10^{14} Hz
(D) 9.0×10^{14} Hz

Che_q008 key **B**

8. Identify the product 'C' formed in the following series of reactions.
Bromoethane $\xrightarrow[\text{Dry ether}]{\text{Mg}}$ A $\xrightarrow[\text{Dry ether}]{\text{HOH}}$ B $\xrightarrow[\text{UV light}]{\text{Br}_2}$ C
(A) Ethane
(B) Bromoethane
(C) Ethyl magnesium bromide
(D) Ethene

Identify the product 'C' formed in the following series of reactions.
$\text{Bromoethane} \xrightarrow[\text{Dry ether}]{\text{Mg}} A \xrightarrow[\text{Dry ether}]{\text{HOH}} B \xrightarrow[\text{UV light}]{\text{Br}_2} C$
(A) Ethane
(B) Bromoethane
(C) Ethyl magnesium bromide
(D) Ethene

Che_q009 key **C**

(D) Ethene

9. Which from following groups exhibits +I effect?

(A) –COOR (B) –COOH

(C) $-C_2H_5$ (D) $-NO_2$

10. Which of the following equation exhibits integrated rate law equation for first order reaction?

(A) $k = \frac{[A]_0 - [A]_t}{2}$

(B) $k = \frac{2 \cdot 303}{t} \times \log \frac{a}{(a-x)}$

6

ents and

(C) $k = \frac{1}{t} \ln \frac{(a-x)}{a}$

(D) $k = \frac{1}{t} \times \frac{x}{a(a-x)}$

3

Which from following groups exhibits $+I$ effect?

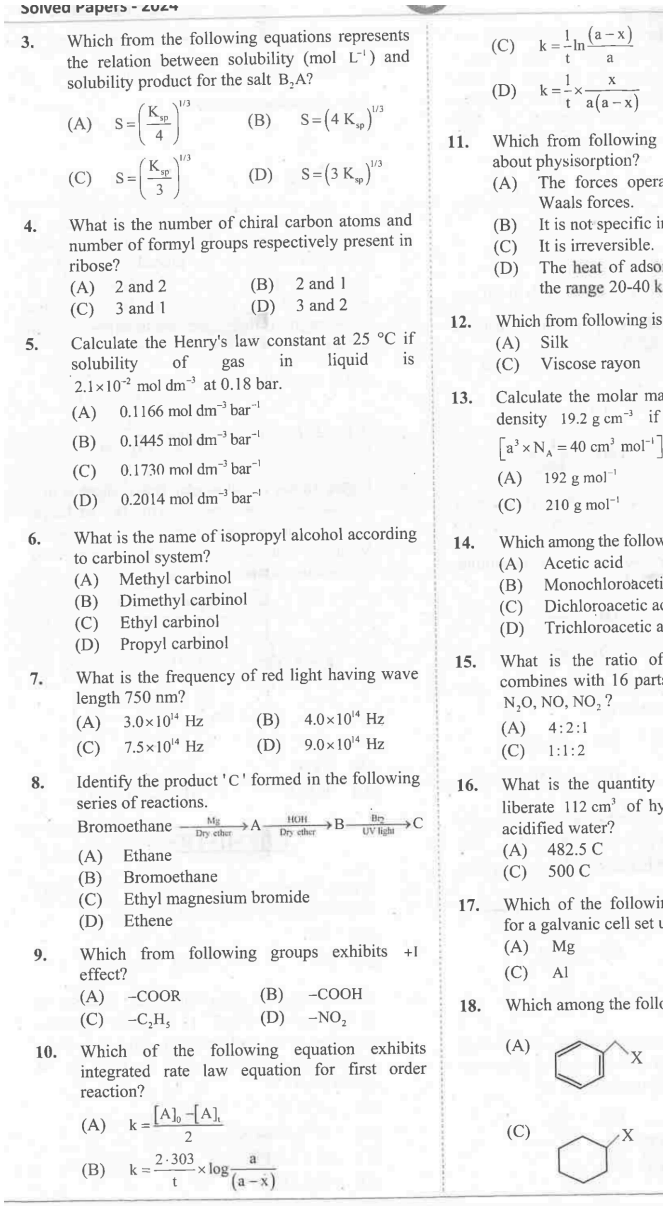
(A) $-\mathrm{COOR}$

(B) $-\mathrm{COOH}$

(C) $-\mathrm{C_2H_5}$

(D) $-\mathrm{NO_2}$

Che_q010 key **B**



Solved Papers - 2024

3. Which from the following equations represents the relation between solubility (mol L^{-1}) and solubility product for the salt B_2A?
(A) $S=\left(\frac{K_{sp}}{4}\right)^{1/3}$ (B) $S=\left(4\,K_{sp}\right)^{1/3}$
(C) $S=\left(\frac{K_{sp}}{3}\right)^{1/3}$ (D) $S=\left(3\,K_{sp}\right)^{1/3}$

4. What is the number of chiral carbon atoms and number of formyl groups respectively present in ribose?
(A) 2 and 2 (B) 2 and 1
(C) 3 and 1 (D) 3 and 2

5. Calculate the Henry's law constant at 25 °C if solubility of gas in liquid is 2.1×10^{-2} mol dm^{-3} at 0.18 bar.
(A) 0.1166 mol dm^{-3} bar^{-1}
(B) 0.1445 mol dm^{-3} bar^{-1}
(C) 0.1730 mol dm^{-3} bar^{-1}
(D) 0.2014 mol dm^{-3} bar^{-1}

6. What is the name of isopropyl alcohol according to carbinol system?
(A) Methyl carbinol
(B) Dimethyl carbinol
(C) Ethyl carbinol
(D) Propyl carbinol

7. What is the frequency of red light having wave length 750 nm?
(A) 3.0×10^{14} Hz (B) 4.0×10^{14} Hz
(C) 7.5×10^{14} Hz (D) 9.0×10^{14} Hz

8. Identify the product 'C' formed in the following series of reactions.
Bromoethane $\xrightarrow[\text{Dry ether}]{Mg}$ A $\xrightarrow[\text{Dry ether}]{HOH}$ B $\xrightarrow[\text{UV light}]{Br_2}$ C
(A) Ethane
(B) Bromoethane
(C) Ethyl magnesium bromide
(D) Ethene

9. Which from following groups exhibits +I effect?
(A) –COOR (B) –COOH
(C) $-C_2H_5$ (D) $-NO_2$

10. Which of the following equation exhibits integrated rate law equation for first order reaction?
(A) $k=\frac{[A]_0-[A]_t}{2}$
(B) $k=\frac{2.303}{t}\times\log\frac{a}{(a-x)}$
(C) $k=\frac{1}{t}\ln\frac{(a-x)}{a}$
(D) $k=\frac{1}{t}\times\frac{x}{a(a-x)}$

11. Which from following about physisorption?
(A) The forces opera Waals forces.
(B) It is not specific i
(C) It is irreversible.
(D) The heat of adso the range 20-40 k

12. Which from following is
(A) Silk
(C) Viscose rayon

13. Calculate the molar ma density 19.2 g cm^{-3} if $\left[a^3\times N_A=40\ cm^3\ mol^{-1}\right]$
(A) 192 g mol^{-1}
(C) 210 g mol^{-1}

14. Which among the follow
(A) Acetic acid
(B) Monochloroaceti
(C) Dichloroacetic a
(D) Trichloroacetic a

15. What is the ratio of combines with 16 part N_2O, NO, NO_2 ?
(A) 4:2:1
(C) 1:1:2

16. What is the quantity liberate 112 cm^3 of hy acidified water?
(A) 482.5 C
(C) 500 C

17. Which of the followi for a galvanic cell set
(A) Mg
(C) Al

18. Which among the foll
(A)
(C)

Which of the following equation exhibits integrated rate law equation for first order reaction?

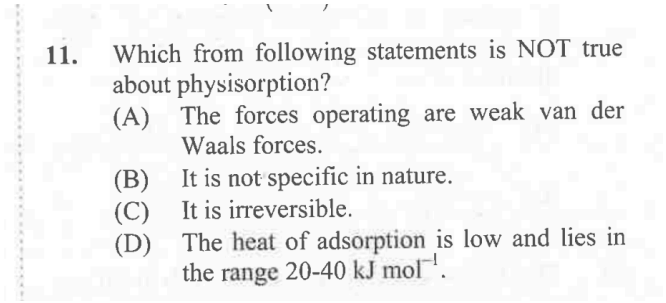
(A) $k=\frac{[A]_0-[A]_t}{2}$

(B) $k=\frac{2.303}{t}\times\log\frac{a}{(a-x)}$

(C) $k=\frac{1}{t}\ln\frac{(a-x)}{a}$

(D) $k=\frac{1}{t}\times\frac{x}{a(a-x)}$

Che_q011 key **C**



11. Which from following statements is NOT true about physisorption?
(A) The forces operating are weak van der Waals forces.
(B) It is not specific in nature.
(C) It is irreversible.
(D) The heat of adsorption is low and lies in the range 20-40 kJ mol^{-1}.

Which from following statements is NOT true about physisorption?

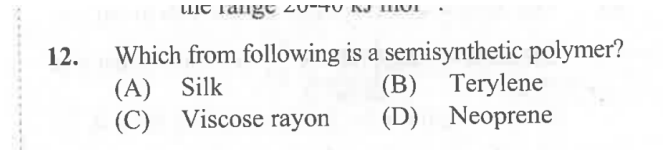
(A) The forces operating are weak van der Waals forces.

(B) It is not specific in nature.

(C) It is irreversible.

(D) The heat of adsorption is low and lies in the range $20-40\,\text{kJ mol}^{-1}$

Che_q012 key **C**



the range 20-40 kJ mol^{-1}.

12. Which from following is a semisynthetic polymer?
(A) Silk (B) Terylene
(C) Viscose rayon (D) Neoprene

Which from following is a semisynthetic polymer?

(A) Silk

(B) Terylene

(C) Viscose rayon

(D) Neoprene

Che_q013 key **A**

13. Calculate the molar mass of an element having density 19.2 g cm^{-3} if it forms fcc structure $\left[a^3 \times N_A = 40\ cm^3\ mol^{-1}\right]$

(A) 192 g mol^{-1} (B) 186 g mol^{-1}

(C) 210 g mol^{-1} (D) 280 g mol^{-1}

Calculate the molar mass of an element having density $19.2\ \text{g cm}^{-3}$ if it forms fcc structure $\left[a^3 \times N_A = 40\ \text{cm}^3\ \text{mol}^{-1}\right]$

(A) $192\ \text{g mol}^{-1}$

(B) $186\ \text{g mol}^{-1}$

(C) $210\ \text{g mol}^{-1}$

(D) $280\ \text{g mol}^{-1}$

Che_q014 key **D**

14. Which among the following is the strongest acid ?

(A) Acetic acid

(B) Monochloroacetic acid

(C) Dichloroacetic acid

(D) Trichloroacetic acid

Which among the following is the strongest acid?

(A) Acetic acid

(B) Monochloroacetic acid

(C) Dichloroacetic acid

(D) Trichloroacetic acid

Che_q015 key **A** OPTION_EMPTY

15. What is the ratio of mass of nitrogen that combines with 16 parts by weight of oxygen in N_2O, NO, NO_2 ?

(A) 4:2:1 (B) 2:1:1

(C) 1:1:2 (D) 1:2:4

What is the ratio of mass of nitrogen that combines with 16 parts by weight of oxygen in N_2O, NO, NO_2?

(A) 4:2:1

(B) 2:1:1

(C) 1:1:2

(D) 1:2:4

Che_q016 key **B**

16. What is the quantity of electricity required to liberate 112 cm^3 of hydrogen gas at STP from acidified water?

(A) 482.5 C (B) 965 C

(C) 500 C (D) 96500 C

What is the quantity of electricity required to liberate $112\ \text{cm}^3$ of hydrogen gas at STP from acidified water?

(A) 482.5 C

(B) 965 C

(C) 500 C

(D) 96500 C

Che_q017 key **B** OPTION_EMPTY

17. Which of the following is selected as cathode for a galvanic cell set up with nickel anode?

(A) Mg (B) Cu

(C) Al (D) Zn

Which of the following is selected as cathode for a galvanic cell set up with nickel anode?

(A) Mg

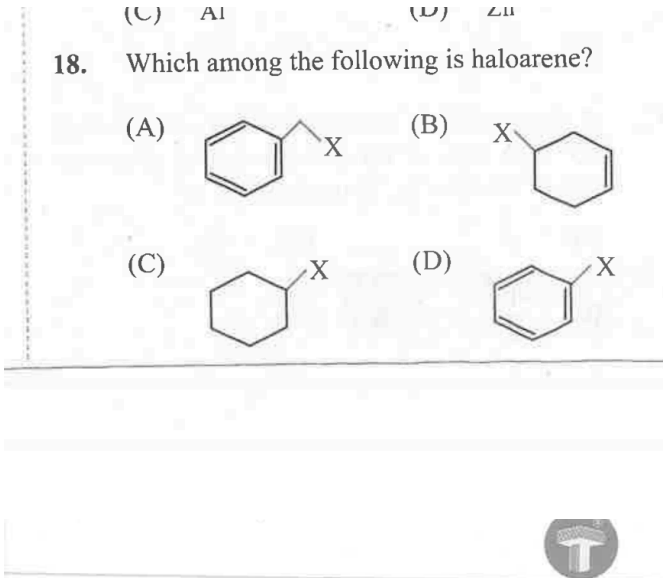
(B) Cu

(C) Al

(D) Zn

Che_q018 key **D** OPTION_EMPTY

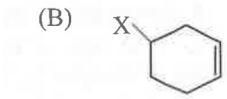
The crop shows a phenyl halide in D (the only aryl halide), but our option text is blank, so the options must be transcribed from the crop before the item is usable.



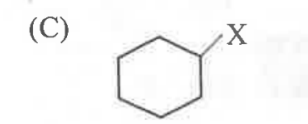
Which among the following is haloarene?

(A)

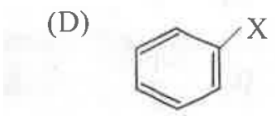
(B)



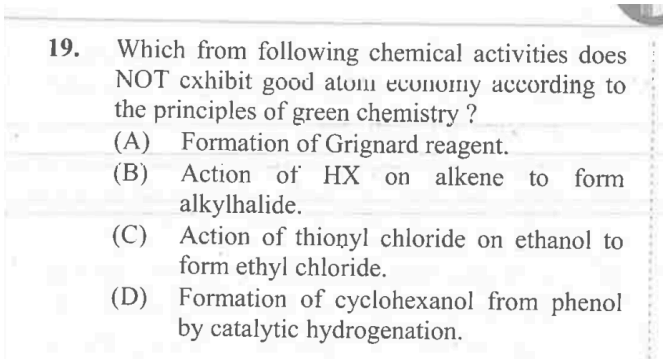
(C)



(D)



Che_q019 key **C**



Which from following chemical activities does NOT exhibit good atom economy according to the principles of green chemistry?

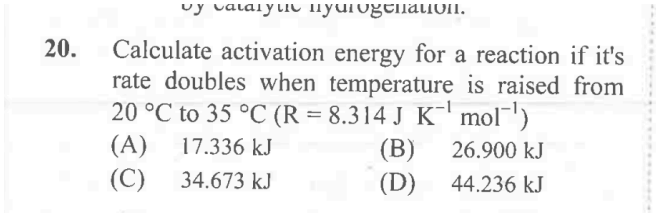
(A) Formation of Grignard reagent.

(B) Action of HX on alkene to form alkylhalide.

(C) Action of thionyl chloride on ethanol to form ethyl chloride.

(D) Formation of cyclohexanol from phenol by catalytic hydrogenation.

Che_q020 key **C**



Calculate activation energy for a reaction if it's rate doubles when temperature is raised from $20^\circ\mathrm{C}$ to $35^\circ\mathrm{C}$ ($R = 8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$)

(A) $17.336\,\mathrm{kJ}$

(B) $26.900\,\mathrm{kJ}$

(C) $34.673\,\mathrm{kJ}$

(D) $44.236\,\mathrm{kJ}$

Che_q021 key **C** V9_NEEDS_HUMAN

The crop shows no O numbering: the left O (lone pair, triple bond) has formal charge +1 (key C), but the right single-bonded O has -1 (layer1's answer); the numbering decides it.

21. What is the formal charge present on oxygen atom (numbered 1) in Lewis structure of CO_2?

$:O \equiv C - \ddot{O}:$

(A) 0 (B) –1
(C) +1 (D) –2

What is the formal charge present on oxygen atom (numbered 1) in Lewis structure of CO_2?

$: O \equiv C - \ddot{O} :$

(A) 0
(B) -1
(C) +1
(D) -2

Che_q022 key **A**

22. Which among the following is a monomer of natural rubber?
(A) 2-Methyl-1, 3-butadiene
(B) 1, 1, 2, 2-Tetrafluoroethene
(C) 2-Chloro-1, 3 butadiene
(D) 1, 3-butadiene

Which among the following is a monomer of natural rubber?
(A) 2-Methyl-1,3-butadiene
(B) 1,1,2,2-Tetrafluoroethene
(C) 2-Chloro-1,3 butadiene
(D) 1,3-butadiene

Che_q023 key **A**

23. Identify the product obtained when alkyl alkanoate is hydrolysed with dilute HCl.
(A) Alkanol and Alkanoic acid
(B) Alkanal and Alkanone
(C) Alkanone and Alkanoic acid
(D) Alkanol and Alkanal

Identify the product obtained when alkyl alkanoate is hydrolysed with dilute HCl.
(A) Alkanol and Alkanoic acid
(B) Alkanal and Alkanone
(C) Alkanone and Alkanoic acid
(D) Alkanol and Alkanal

Che_q024 key **A**

24. Which of the following changes exhibit that nitrogen undergoes oxidation?
(A) $NH_4^+ \rightarrow N_2$ (B) $NO_3^- \rightarrow NO$
(C) $NO_2 \rightarrow NO_2^-$ (D) $NO_3 \rightarrow NH_4^+$

Which of the following changes exhibit that nitrogen undergoes oxidation?
(A) $NH_4^+ \rightarrow N_2$
(B) $NO_3^- \rightarrow NO$
(C) $NO_2 \rightarrow NO_2^-$
(D) $NO_3^- \rightarrow NH_4^+$

Che_q025 key **C**

25. The molar conductivity of 0.02 mol dm^{-3} solution of sodium hydroxide is 230.5 Ω^{-1} cm^2 mol^{-1}. What is it's conductivity?
(A) 0.01155 Ω^{-1} cm^{-1}
(B) 0.02308 Ω^{-1} cm^{-1}
(C) 0.00461 Ω^{-1} cm^{-1}
(D) 0.05613 Ω^{-1} cm^{-1}

The molar conductivity of $0.02\ \text{mol dm}^{-3}$ solution of sodium hydroxide is $230.5\ \Omega^{-1}\ \text{cm}^2\ \text{mol}^{-1}$. What is it's conductivity?
(A) $0.01155\ \Omega^{-1}\ \text{cm}^{-1}$
(B) $0.02308\ \Omega^{-1}\ \text{cm}^{-1}$
(C) $0.00461\ \Omega^{-1}\ \text{cm}^{-1}$
(D) $0.05613\ \Omega^{-1}\ \text{cm}^{-1}$

Che_q026 key **A**

26. Calculate the volume occupied by particle in simple cubic unit cell if volume of unit cell is 5.5×10^{-22} cm^3.

(A) 2.88×10^{-22} cm^3 (B) 1.87×10^{-22} cm^3
(C) 1.02×10^{-22} cm^3 (D) 3.44×10^{-22} cm^3

Calculate the volume occupied by particle in simple cubic unit cell if volume of unit cell is 5.5×10^{-22} cm^3.

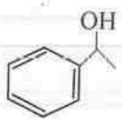
(A) 2.88×10^{-22} cm^3
(B) 1.87×10^{-22} cm^3
(C) 1.02×10^{-22} cm^3
(D) 3.44×10^{-22} cm^3

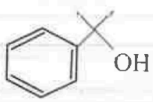
Che_q027 key **C** OPTION_EMPTY

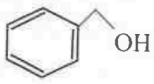
The crop shows benzyl alcohol in C (CH2OH on benzene), the only primary benzylic alcohol, but our option text is blank and must be transcribed from the crop.

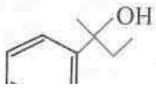
22nd April (Shift – I)

27. Which of the following is primary benzylic alcohol?

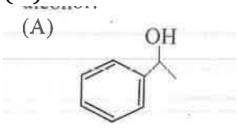
(A) 

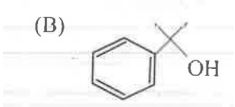
(B) 

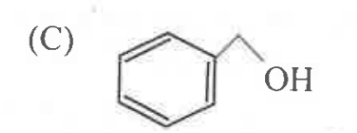
(C) 

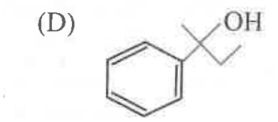
(D) 

Which of the following is primary benzylic alcohol?

(A) 

(B) 

(C) 

(D) 

Che_q028 key **C**

28. Which from following elements is a decay product of thorium?

(A) S (B) Se
(C) Po (D) Te

Which from following elements is a decay product of thorium?

(A) S
(B) Se
(C) Po
(D) Te

Che_q029 key **B**

29. If half life of a first order reaction is 10 minutes, find the time required to decrease concentration of reactant from 0.08 M to 0.02 M.

(A) 10 minutes (B) 20 minutes
(C) 30 minutes (D) 40 minutes

If half life of a first order reaction is 10 minutes, find the time required to decrease concentration of reactant from 0.08 M to 0.02 M.

(A) 10 minutes
(B) 20 minutes
(C) 30 minutes
(D) 40 minutes

Che_q030 key **C**

(C) 50 minutes (D) 40 minutes

30. Calculate the pOH of buffer solution formed from 0.3 M weak base and 0.45 M of its salt with strong acid $[pK_b = 4.7447]$

(A) 6.45 (B) 4.07
(C) 4.92 (D) 5.51

Calculate the pOH of buffer solution formed from $0.3\ M$ weak base and $0.45\ M$ of its salt with strong acid $[pK_b = 4.7447]$

(A) 6.45
(B) 4.07
(C) 4.92
(D) 5.51

Che_q031 key **B**

(C) 4.92 (D) 5.51

31. Which from following complexes is having ambidentate ligand in it?

(A) Sodium hexafluoroaluminate(III)
(B) Pentaaquaisothiocyanatoiron(III) ion
(C) Hexacyanoferrate(II) ion
(D) Trioxalatocobaltate(III) ion

Which from following complexes is having ambidentate ligand in it?

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(D) Trioxalatocobaltate(III) ion

Che_q032 key **C**

(D) Trioxalatocobaltate(III) ion

32. What is the minimum number of spheres required to develop an octahedral void?

(A) 2 (B) 4
(C) 6 (D) 8

What is the minimum number of spheres required to develop an octahedral void?

(A) 2
(B) 4
(C) 6
(D) 8

Che_q033 key **B**

(C) 6 (D) 8

33. How many moles of iodomethane are consumed in the following conversion?

$CH_3NH_2 \xrightarrow[\Delta]{CH_3I} (CH_3)_4N^+I^-$

(A) Four (B) Three
(C) Two (D) One

How many moles of iodomethane are consumed in the following conversion?

$CH_3NH_2 \xrightarrow[\Delta]{CH_3I} (CH_3)_4N^+I^-$

(A) Four
(B) Three
(C) Two
(D) One

Che_q034 key **D**

(C) Two (D) One

34. Identify the use of calcium carbonate from following.

(A) For mercerising cotton fabrics.
(B) As a source of hydride.
(C) As chlorinating agent.
(D) In tooth paste.

Identify the use of calcium carbonate from following.

(A) For mercerising cotton fabrics.
(B) As a source of hydride.
(C) As chlorinating agent.
(D) In tooth paste.

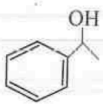
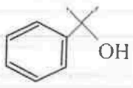
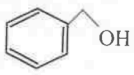
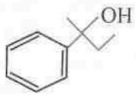
Che_q035 key **D**

(D) In tooth paste.

35. Which of the following acid strength is used to get good yield of alkyl iodide from alcohol and sodium iodide in phosphoric acid?
(A) 15% (B) 35%
(C) 65% (D) 95%

7

MHT - CET 2024
22nd April (Shift – I)

Which of the following is primary benzylic alcohol?
(A)  (B) 
(C)  (D) 

Which from following elements is a decay product of thorium?
(A) S (B) Se
(C) Po (D) Te

Which of the following acid strength is used to get good yield of alkyl iodide from alcohol and sodium iodide in phosphoric acid?

(A) 15%

(B) 35%

(C) 65%

(D) 95%

Che_q036 key **B**

Solved Papers - 2024

36. Identify false statement about transition elements.
(A) As per IUPAC convention, transition metal atom have incomplete d-subshell.
(B) 6d-series includes all elements from $Ac(Z=89)$ to $Cn(Z=112)$.
(C) These elements are placed in group 3 to 12 in modern periodic table.
(D) In these (n – 1) d-orbital is filled successively.

Identify false statement about transition elements.

(A) As per IUPAC convention, transition metal atom have incomplete d-subshell.

(B) 6d-series includes all elements from $\mathrm{Ac}(Z = 89)$ to $\mathrm{Cn}(Z = 112)$.

(C) These elements are placed in group 3 to 12 in modern periodic table.

(D) In these $(n-1)$ d-orbital is filled successively.

Che_q037 key **C**

37. Which from following terms is explained by first law of thermodynamics?
(A) Entropy
(B) Free energy
(C) Conservation of energy
(D) Enthalpy

Which from following terms is explained by first law of thermodynamics?

(A) Entropy

(B) Free energy

(C) Conservation of energy

(D) Enthalpy

Che_q038 key **C**

38. Calculate degree of dissociation of a weak monobasic acid in 0.01 M solution if dissociation constant is 1.6×10^{-5}.
(A) 0.02 (B) 0.03
(C) 0.04 (D) 0.05

Calculate degree of dissociation of a weak monobasic acid in 0.01 M solution if dissociation constant is 1.6×10^{-5}.
(A) 0.02
(B) 0.03
(C) 0.04
(D) 0.05

Che_q039 key **A**

39. Which from following is an example of solution of solid as solute and liquid as solvent?
(A) Sea water (B) Brass
(C) Gasoline (D) Iodine in air

Which from following is an example of solution of solid as solute and liquid as solvent?
(A) Sea water
(B) Brass
(C) Gasoline
(D) Iodine in air

Che_q040 key **C** SOLVER_DISAGREES_WITH_BOOK, V9_KEY_DOUBT

Al is bound by three bidentate oxalate ligands, giving coordination number 6 (C); book key D (12) is wrong.

40. Identify the coordination number of aluminium in potassium trioxalatoaluminate(III).
(A) 2 (B) 4
(C) 6 (D) 12

Identify the coordination number of aluminium in potassium trioxalatoaluminate(III).
(A) 2
(B) 4
(C) 6
(D) 12

Che_q041 key **D**

41. Which of the following isomeric amines has the highest boiling point?
(A) tert-Butylamine
(B) Ethyldimethylamine
(C) Diethylamine
(D) n-Butylamine

Which of the following isomeric amines has the highest boiling point?
(A) tert-Butylamine
(B) Ethyldimethylamine
(C) Diethylamine
(D) n-Butylamine

Che_q042 key **B**

42. If A, B, C, D are four different elements with outer electronic configuration as $A = 4s^2 4p^4$, $B = 4s^2 4p^5$, $C = 5s^2 5p^4$, $D = 5s^2 5p^5$.
Find the element having highest ionization enthalpy $(\Delta_i H_1)$
(A) A (B) B
(C) C (D) D

If A, B, C, D are four different elements with outer electronic configuration as $A = 4s^2 4p^4$, $B = 4s^2 4p^5$, $C = 5s^2 5p^4$, $D = 5s^2 5p^5$. Find the element having highest ionization enthalpy $(\Delta_i H_1)$
(A) A
(B) B
(C) C
(D) D

Che_q043 key **A**

43. What is the change in internal energy for $2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$ at 25 °C ?
($R = 8.314\ J\ K^{-1}\ mol^{-1}$, $\Delta H = -560\ kJ$)
(A) −557.5 kJ (B) −530.0 kJ
(C) 510.0 kJ (D) 656.9 kJ

8

What is the change in internal energy for $2\,CO_{(g)} + O_{2(g)} \rightarrow 2\,CO_{2(g)}$ at $25^\circ C$?
$(R = 8.314\,J\,K^{-1}\,mol^{-1}, \Delta H = -560\,kJ)$
(A) $-557.5\,kJ$
(B) $-530.0\,kJ$
(C) $510.0\,kJ$
(D) $656.9\,kJ$

Che_q044 key **D**

44. Calculate the final pressure required to reduce the volume of a gas to one third, if initial pressure is $1.6\times10^5\ Nm^{-2}$.
(A) $5.4\times10^5\ Nm^{-2}$ (B) $1.6\times10^5\ Nm^{-2}$
(C) $2.11\times10^5\ Nm^{-2}$ (D) $4.8\times10^5\ Nm^{-2}$

Calculate the final pressure required to reduce the volume of a gas to one third, if initial pressure is $1.6 \times 10^5\,N\,m^{-2}$.
(A) $5.4 \times 10^5\,N\,m^{-2}$
(B) $1.6 \times 10^5\,N\,m^{-2}$
(C) $2.11 \times 10^5\,N\,m^{-2}$
(D) $4.8 \times 10^5\,N\,m^{-2}$

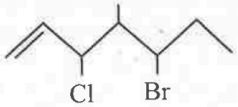
Che_q045 key **A**

45. If osmotic pressure of 0.2 M aqueous glucose solution is 5 atm at 300 K. Calculate the concentration of glucose solution having osmotic pressure 2.5 atm at same temperature.
(A) 0.1 M (B) 0.05 M
(C) 0.75 M (D) 0.25 M

If osmotic pressure of $0.2\,M$ aqueous glucose solution is $5\,atm$ at $300\,K$. Calculate the concentration of glucose solution having osmotic pressure $2.5\,atm$ at same temperature.
(A) $0.1\,M$
(B) $0.05\,M$
(C) $0.75\,M$
(D) $0.25\,M$

Che_q046 key **D**

46. What is the IUPAC name of following compound?



(A) 3-Chloro-4-methyl-5-bromohept-1-ene
(B) 4-Methyl-3-bromo-5-chlorohept-6-ene
(C) 3-Bromo-4-methyl-5-chlorohept-6-ene
(D) 5-Bromo-3-chloro-4-methylhept-1-ene

What is the IUPAC name of following compound? [figure]
(A) 3-Chloro-4-methyl-5-bromohept-1-ene
(B) 4-Methyl-3-bromo-5-chlorohept-6-ene
(C) 3-Bromo-4-methyl-5-chlorohept-6-ene
(D) 5-Bromo-3-chloro-4-methylhept-1-ene

Che_q047 key **C**

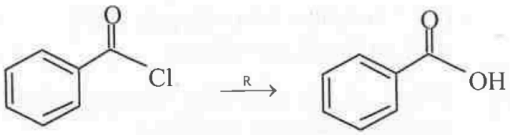
47. Which amino acid from following contains highest number of N atoms in it?

(A) Histidine (B) Lysine
(C) Arginine (D) Glutamine

Which amino acid from following contains highest number of N atoms in it?
(A) Histidine
(B) Lysine
(C) Arginine
(D) Glutamine

Che_q048 key **A**

48. Identify the reagent 'R' necessary to bring the following conversion.



(A) H_2O (B) CrO_3
(C) NaOH (dil.) (D) $KMnO_4$

Identify the reagent 'R' necessary to bring the following conversion. [figure]
(A) H_2O
(B) CrO_3
(C) $NaOH$ (dil.)
(D) $KMnO_4$

Che_q049 key **C**

49. Identify 'A' in the following reaction.

$$A + \text{Acetic anhydride} \xrightarrow{H^{\oplus}} \text{Aspirin} + \text{Acetic acid}$$

(A) Acrylic acid
(B) Oxalic acid
(C) Salicylic acid
(D) Phthalic acid

Identify 'A' in the following reaction.

$$A + \text{Acetic anhydride} \xrightarrow{H^{+}} \text{Aspirin} + \text{Acetic acid}$$

(A) Acrylic acid
(B) Oxalic acid
(C) Salicylic acid
(D) Phthalic acid

Che_q050 key **B**

50. What are the possible values of magnetic quantum number for p orbital?

(A) 2 (B) 3
(C) 5 (D) 6

Biology

Time: 90 Minutes

What are the possible values of magnetic quantum number for p orbital?
(A) 2
(B) 3
(C) 5
(D) 6

Bio_q001 key **D**

1. In eukaryotic organisms, replication of DNA occurs in _____.
(A) anaphase (B) G_1 phase
(C) G_2 phase (D) S phase

In eukaryotic organisms, replication of DNA occurs in _____.
(A) anaphase
(B) G_1 phase
(C) G_2 phase
(D) S phase

Bio_q002 key **A**

2. Capillarity theory was proposed by_____
(A) Bohem (B) J. Priestley
(C) Dixon and Joly (D) Levitt

Capillarity theory was proposed by _____.
(A) Bohem
(B) J. Priestley
(C) Dixon and Joly
(D) Levitt

Bio_q003 key **A**

3. Given below are two statements.
Statement I - Oxaloacetate is useful as a precursor for synthesis of aspartic acid.
Statement II - α-ketoglutarate is useful for synthesis of glutamic acid.
In light of above statements, select the correct answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Given below are two statements.
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(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q004 key **D**

4. Given below are two statements.
Statement I - Defecation is an involuntary process that takes place through anal opening guarded by sphincter muscles.
Statement II - Distension of rectum stimulates pressure sensitive receptors that initiate a neural reflex for egestion.
In light of above statements, select the correct answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

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In light of above statements, select the correct answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q005 key **C**

5. Angina pectoris is severe pain and heaviness in chest that is mainly caused due to
   (A) metabolic disorder of pancreas
   (B) deficiency of iodine in diet
   (C) reduction in blood supply to cardiac muscles due to atherosclerosis
   (D) long term exposure to dust particles as in silicosis

Angina pectoris is severe pain and heaviness in chest that is mainly caused due to
(A) metabolic disorder of pancreas
(B) deficiency of iodine in diet
(C) reduction in blood supply to cardiac muscles due to atherosclerosis
(D) long term exposure to dust particles as in silicosis

Bio_q006 key **B**

6. The term 'Niche' denotes the_____ role played by an organism in its environment.
   (A) chemical (B) functional
   (C) biological (D) physical

The term 'Niche' denotes the _____ role played by an organism in its environment.
(A) chemical
(B) functional
(C) biological
(D) physical

Bio_q007 key **B**

7. When the phage DNA enters the host bacterium, the host cell protects itself from the viral DNA attacks with the help of _____ enzyme.
   (A) exonuclease
   (B) restriction endonuclease
   (C) DNA ligase
   (D) helicase

When the phage DNA enters the host bacterium, the host cell protects itself from the viral DNA attacks with the help of _____ enzyme.
(A) exonuclease
(B) restriction endonuclease
(C) DNA ligase
(D) helicase

Bio_q008 key **C**

8. Given below are two statements.
   Statement I - Nervous tissue is without lymphatic vessels.
   Statement II - Nervous tissue is endodermal in origin.
   In light of above statements, choose the most appropriate correct answer from the option given below.
   (A) Both statement I and statement II are correct.
   (B) Both statement I and statement II are incorrect.
   (C) Statement I is correct and statement II is incorrect.
   (D) Statement I is incorrect and statement II is correct.

Given below are two statements.
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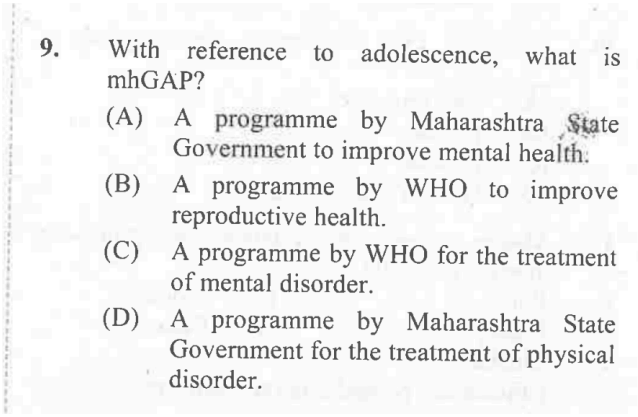
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q009 key **C**



9. With reference to adolescence, what is mhGAP?
   (A) A programme by Maharashtra State Government to improve mental health.
   (B) A programme by WHO to improve reproductive health.
   (C) A programme by WHO for the treatment of mental disorder.
   (D) A programme by Maharashtra State Government for the treatment of physical disorder.

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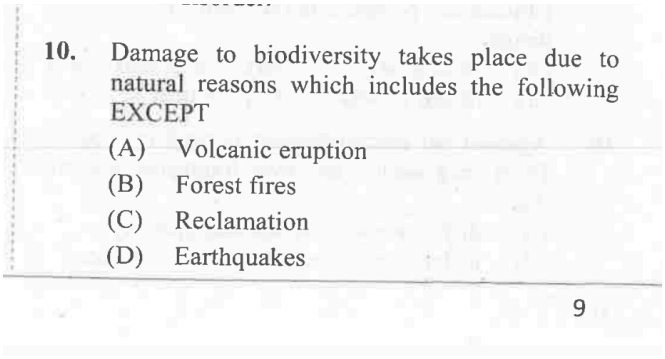
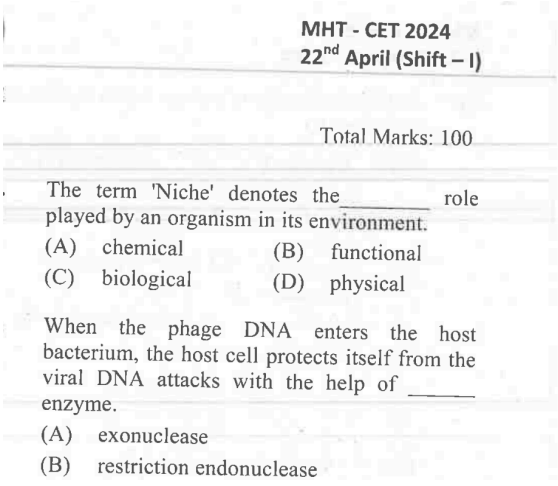
Bio_q010 key **C**



10. Damage to biodiversity takes place due to natural reasons which includes the following EXCEPT
    (A) Volcanic eruption
    (B) Forest fires
    (C) Reclamation
    (D) Earthquakes

9



MHT - CET 2024
22nd April (Shift – I)

Total Marks: 100

The term 'Niche' denotes the_____ role played by an organism in its environment.
(A) chemical (B) functional
(C) biological (D) physical

When the phage DNA enters the host bacterium, the host cell protects itself from the viral DNA attacks with the help of _____ enzyme.
(A) exonuclease
(B) restriction endonuclease

Damage to biodiversity takes place due to natural reasons which includes the following EXCEPT
(A) Volcanic eruption
(B) Forest fires
(C) Reclamation
(D) Earthquakes

Bio_q011 key **D**

Solved Papers - 2024

11. Match the disorder in column I with their nature in column II.

	Column I		Column II
i.	Thalassemia	a.	X monosomy
ii.	Turner's syndrome	b.	Extra X chromosomes in male
iii.	Klinefelter's syndrome	c.	21st Trisomy
iv.	Down's syndrome	d.	Mendelian disorder

(A) i - a ii - c iii - d iv - b
(B) i - b ii - a iii - c iv - d
(C) i - d ii - c iii - a iv - b
(D) i - d ii - a iii - b iv - c

Match the disorder in column I with their nature in column II.
Column I:
i. Thalassemia |
ii. Turner's syndrome |
iii. Klinefelter's syndrome |
iv. Down's syndrome
Column II: a. X monosomy | b. Extra X chromosomes in male | c. 21st Trisomy | d. Mendelian disorder
(A) i-a ii-c iii-d iv-b
(B) i-b ii-a iii-c iv-d
(C) i-d ii-c iii-a iv-b
(D) i-d ii-a iii-b iv-c

Bio_q012 key **B**

12. Chemical digestion of which one of the following starts first in human alimentary canal?
(A) Proteins (B) Starch
(C) Lipids and fats (D) Cellulose

Chemical digestion of which one of the following starts first in human alimentary canal?
(A) Proteins
(B) Starch
(C) Lipids and fats
(D) Cellulose

Bio_q013 key **C**

13. DDT resistant mosquitoes and different beak sizes of African seed cracker finches show _____ type of selection respectively.
(A) stabilizing and directional
(B) disruptive and balancing
(C) directional and disruptive
(D) balancing and directional

DDT resistant mosquitoes and different beak sizes of African seed cracker finches show _____ type of selection respectively.
(A) stabilizing and directional
(B) disruptive and balancing
(C) directional and disruptive
(D) balancing and directional

Bio_q014 key **D**

14. The number of ATP molecules generated in aerobic respiration after oxidation of two molecules of pyruvic acid is.
(A) 6 (B) 12 (C) 24 (D) 30

The number of ATP molecules generated in aerobic respiration after oxidation of two molecules of pyruvic acid is.
(A) 6
(B) 12
(C) 24
(D) 30

Bio_q015 key **D**

15. During oogenesis the meiotic division is arrested at _____ before fertilization.
(A) Anaphase I (B) Metaphase I
(C) Anaphase II (D) Metaphase II

During oogenesis the meiotic division is arrested at _____ before fertilization.
(A) Anaphase I
(B) Metaphase I
(C) Anaphase II
(D) Metaphase II

Bio_q016 key **C**

(C) Anaphase II (D) Metaphase II

16. Accessory cells of stomata are reservoirs of _____.
(A) Zn^{++} ions
(B) H^+ ions
(C) K^+ ions
(D) H^+ ions and Cl^- ions

Accessory cells of stomata are reservoirs of _____.
(A) Zn^{++} ions
(B) H^+ ions
(C) K^+ ions
(D) H^+ ions and Cl^- ions

Bio_q017 key **C**

(D) H^+ ions and Cl^- ions

17. Mature erythrocytes are enucleated in following animals EXCEPT
i. Rat ii. Monkey
iii. Llama iv. Camel
v. Elephant
Choose the correct answer from options given below.
(A) i and ii only (B) iii, iv and v only
(C) iii and iv only (D) v only

18. Agarose gel electrophoresis is used to separate DNA fragments. The DNA fragments separate due to _____.
(A) difference in their staining property
(B) difference in the base sequence in them

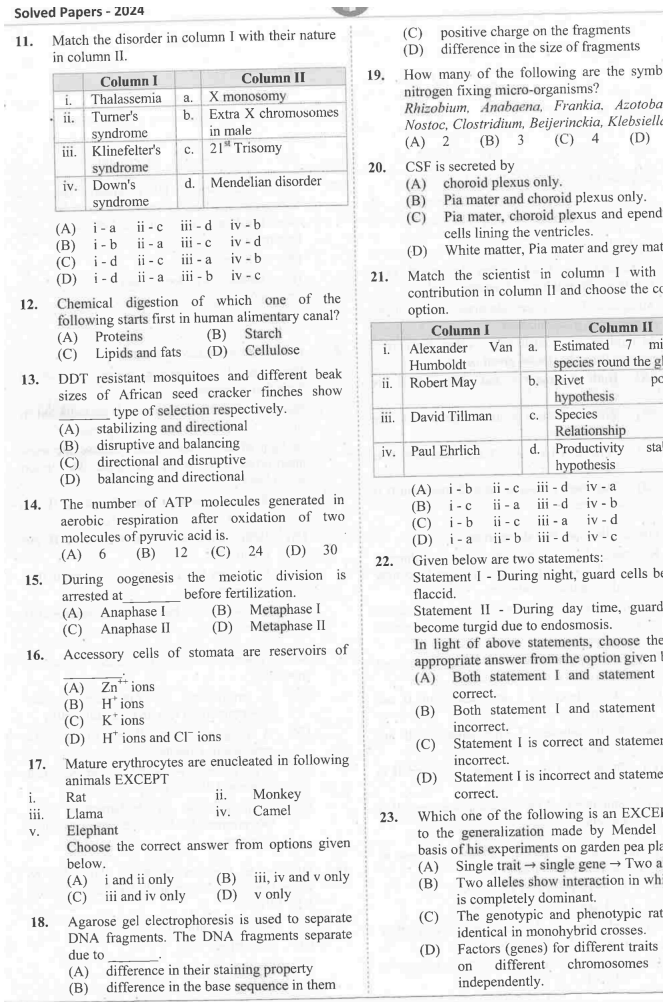
10

ature

(C) positive charge on the fragments
(D) difference in the size of fragments

Mature erythrocytes are enucleated in following animals EXCEPT
i. Rat
ii. Monkey
iii. Llama
iv. Camel
v. Elephant Choose the correct answer from options given below.
(A) i and ii only
(B) iii, iv and v only
(C) iii and iv only
(D) v only

Bio_q018 key **D**



Solved Papers - 2024

11. Match the disorder in column I with their nature in column II.

	Column I		Column II
i.	Thalassemia	a.	X monosomy
ii.	Turner's syndrome	b.	Extra X chromosomes in male
iii.	Klinefelter's syndrome	c.	21st Trisomy
iv.	Down's syndrome	d.	Mendelian disorder

(A) i - a ii - c iii - d iv - b
(B) i - b ii - a iii - c iv - d
(C) i - d ii - c iii - a iv - b
(D) i - d ii - a iii - b iv - c

12. Chemical digestion of which one of the following starts first in human alimentary canal?
(A) Proteins (B) Starch
(C) Lipids and fats (D) Cellulose

13. DDT resistant mosquitoes and different beak sizes of African seed cracker finches show _____ type of selection respectively.
(A) stabilizing and directional
(B) disruptive and balancing
(C) directional and disruptive
(D) balancing and directional

14. The number of ATP molecules generated in aerobic respiration after oxidation of two molecules of pyruvic acid is.
(A) 6 (B) 12 (C) 24 (D) 30

15. During oogenesis the meiotic division is arrested at_____ before fertilization.
(A) Anaphase I (B) Metaphase I
(C) Anaphase II (D) Metaphase II

16. Accessory cells of stomata are reservoirs of _____.
(A) Zn^{++} ions
(B) H^+ ions
(C) K^+ ions
(D) H^+ ions and Cl^- ions

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i. Rat ii. Monkey
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(A) i and ii only (B) iii, iv and v only
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18. Agarose gel electrophoresis is used to separate DNA fragments. The DNA fragments separate due to _____.
(A) difference in their staining property
(B) difference in the base sequence in them
(C) positive charge on the fragments
(D) difference in the size of fragments

19. How many of the following are the symb nitrogen fixing micro-organisms?
Rhizobium, Anabaena, Frankia, Azotoba Nostoc, Clostridium, Beijerinckia, Klebsiell
(A) 2 (B) 3 (C) 4 (D)

20. CSF is secreted by
(A) choroid plexus only.
(B) Pia mater and choroid plexus only.
(C) Pia mater, choroid plexus and epend cells lining the ventricles.
(D) White matter, Pia mater and grey mat

21. Match the scientist in column I with contribution in column II and choose the co option.

	Column I		Column II
i.	Alexander Van Humboldt	a.	Estimated 7 mi species round the gl
ii.	Robert May	b.	Rivet pc hypothesis
iii.	David Tillman	c.	Species Relationship
iv.	Paul Ehrlich	d.	Productivity sta hypothesis

(A) i - b ii - c iii - d iv - a
(B) i - c ii - a iii - d iv - b
(C) i - b ii - c iii - a iv - d
(D) i - a ii - b iii - d iv - c

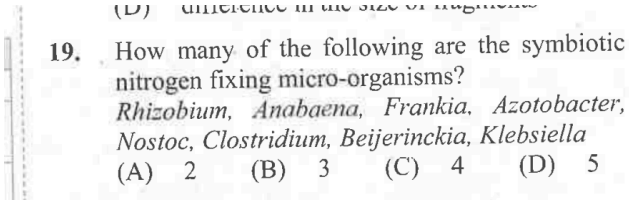
22. Given below are two statements:
Statement I - During night, guard cells be flaccid.
Statement II - During day time, guard become turgid due to endosmosis.
In light of above statements, choose the appropriate answer from the option given l
(A) Both statement I and statement correct.
(B) Both statement I and statement incorrect.
(C) Statement I is correct and statemer incorrect.
(D) Statement I is incorrect and stateme correct.

23. Which one of the following is an EXCEI to the generalization made by Mendel basis of his experiments on garden pea pla
(A) Single trait → single gene → Two a
(B) Two alleles show interaction in wh is completely dominant.
(C) The genotypic and phenotypic rat identical in monohybrid crosses.
(D) Factors (genes) for different traits on different chromosomes independently.

Agarose gel electrophoresis is used to separate DNA fragments. The DNA fragments separate due to _____.
(A) difference in their staining property
(B) difference in the base sequence in them
(C) positive charge on the fragments
(D) difference in the size of fragments

Bio_q019 key **B** SOLVER_DISAGREES_WITH_BOOK, V9_KEY_DOUBT

Symbiotic N-fixers in the list are Rhizobium, Frankia, Anabaena and Nostoc (4 = C); key B (3) would need one of these to be excluded, which the question does not show.

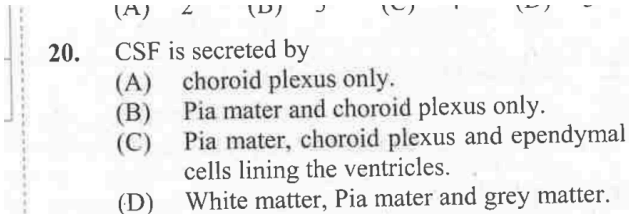


(D) difference in the size of fragments

19. How many of the following are the symbiotic nitrogen fixing micro-organisms?
Rhizobium, Anabaena, Frankia, Azotobacter, Nostoc, Clostridium, Beijerinckia, Klebsiella
(A) 2 (B) 3 (C) 4 (D) 5

How many of the following are the symbiotic nitrogen fixing micro-organisms? Rhizobium, Anabaena, Frankia, Azotobacter, Nostoc, Clostridium, Beijerinckia, Klebsiella
(A) 2
(B) 3
(C) 4
(D) 5

Bio_q020 key **C**



(A) 2 (B) 3 (C) 4 (D)

20. CSF is secreted by
(A) choroid plexus only.
(B) Pia mater and choroid plexus only.
(C) Pia mater, choroid plexus and ependymal cells lining the ventricles.
(D) White matter, Pia mater and grey matter.

CSF is secreted by
(A) choroid plexus only.
(B) Pia mater and choroid plexus only.
(C) Pia mater, choroid plexus and ependymal cells lining the ventricles.
(D) White matter, Pia mater and grey matter.

Bio_q021 key **B**

(D) White matter, Pia mater and grey matter.

21. Match the scientist in column I with their contribution in column II and choose the correct option.

	Column I		Column II
i.	Alexander Van Humboldt	a.	Estimated 7 million species round the globe
ii.	Robert May	b.	Rivet popper hypothesis
iii.	David Tillman	c.	Species Area Relationship
iv.	Paul Ehrlich	d.	Productivity stability hypothesis

(A) i - b ii - c iii - d iv - a
(B) i - c ii - a iii - d iv - b
(C) i - b ii - c iii - a iv - d
(D) i - a ii - b iii - d iv - c

Match the scientist in column I with their contribution in column II and choose the correct option.
Column I:
i. Alexander Van Humboldt |
ii. Robert May |
iii. David Tillman |
iv. Paul Ehrlich
Column II: a. Estimated 7 million species round the globe | b. Rivet popper hypothesis | c. Species Area Relationship | d. Productivity stability hypothesis
(A) i-b ii-c iii-d iv-a
(B) i-c ii-a iii-d iv-b
(C) i-b ii-c iii-a iv-d
(D) i-a ii-b iii-d iv-c

Bio_q022 key **A**

(D) i - a ii - b iii - d iv - c

22. Given below are two statements:
Statement I - During night, guard cells become flaccid.
Statement II - During day time, guard cells become turgid due to endosmosis.
In light of above statements, choose the most appropriate answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Given below are two statements: Statement I - During night, guard cells become flaccid. Statement II - During day time, guard cells become turgid due to endosmosis. In light of above statements, choose the most appropriate answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q023 key **C**

correct.

23. Which one of the following is an EXCEPTION to the generalization made by Mendel on the basis of his experiments on garden pea plant?
(A) Single trait → single gene → Two alleles.
(B) Two alleles show interaction in which one is completely dominant.
(C) The genotypic and phenotypic ratios are identical in monohybrid crosses.
(D) Factors (genes) for different traits present on different chromosomes assort independently.

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(B) Two alleles show interaction in which one is completely dominant.
(C) The genotypic and phenotypic ratios are identical in monohybrid crosses.
(D) Factors (genes) for different traits present on different chromosomes assort independently.

Bio_q024 key **B**

24. People who consume high protein diet can develop _____ stones.
(A) calcium (B) uric acid
(C) stuvite (D) cystine

People who consume high protein diet can develop _____ stones.
(A) calcium
(B) uric acid
(C) stuvite
(D) cystine

Bio_q025 key **A**

(C) stuvite (D) cystine

25. In an angiospermic embryo, upper swollen suspensor cell towards the micropyle, functions as _____.
(A) haustorium (B) hypophysis
(C) epicotyl (D) hypocotyl

In an angiospermic embryo, upper swollen suspensor cell towards the micropyle, functions as _____.
(A) haustorium
(B) hypophysis
(C) epicotyl
(D) hypocotyl

Bio_q026 key **C**

(C) epicotyl (D) hypocotyl

26. In large proteins, peptide chains are much looped, twisted and folded back on themselves. The folded structure is due to the formation of _____ bonds.
(A) phosphodiester (B) peptide
(C) disulphide (D) glycosidic

In large proteins, peptide chains are much looped, twisted and folded back on themselves. The folded structure is due to the formation of _____ bonds.
(A) phosphodiester
(B) peptide
(C) disulphide
(D) glycosidic

Bio_q027 key **D**

(C) disulphide (D) glycosidic

27. Arrange the following in sequence during the transmission of nerve impulse at chemical synapse.
i. Neurotransmitter binds with receptors of post synaptic cell.
ii. Calcium channels open and the calcium diffuses inward from ECF.
iii. Impulse travels along the axon of presynaptic neuron.
iv. Release of neurotransmitter by exocytosis.
v. Synaptic vesicles fuse with cell membrane of axon terminal of presynaptic neuron.
(A) i, ii, v, iii, iv (B) iii, v, ii, iv, i
(C) ii, v, i, iii, iv (D) iii, ii, v, iv, i

Arrange the following in sequence during the transmission of nerve impulse at chemical synapse.

i. Neurotransmitter binds with receptors of post synaptic cell.
ii. Calcium channels open and the calcium diffuses inward from ECF.
iii. Impulse travels along the axon of presynaptic neuron.
iv. Release of neurotransmitter by exocytosis.
v. Synaptic vesicles fuse with cell membrane of axon terminal of presynaptic neuron.
(A) i, ii, v, iii, iv
(B) iii, v, ii, iv, i
(C) ii, v, i, iii, iv
(D) iii, ii, v, iv, i

Bio_q028 key **A**

(C) ii, v, i, iii, iv (D) iii, ii, v, iv, i

28. The highest level of ecological hierarchy in ecological organization is
(A) biome (B) community
(C) population (D) organism

The highest level of ecological hierarchy in ecological organization is
(A) biome
(B) community
(C) population
(D) organism

Bio_q029 key **D**

(C) population (D) organism

29. Primary productivity in an ecosystem refers to the rate of generation of _____.
(A) oxygen (B) carbon dioxide
(C) detritus (D) biomass

Primary productivity in an ecosystem refers to the rate of generation of _____.
(A) oxygen
(B) carbon dioxide
(C) detritus
(D) biomass

Bio_q030 key **A**

(C) detritus (D) biomass

30. Cells of malignant tumor spread from one organ to other via blood or lymph and cause new tumor. This property is called _____.
(A) metastasis (B) parthenogenesis
(C) interkinesis (D) metamorphosis

Cells of malignant tumor spread from one organ to other via blood or lymph and cause new tumor. This property is called _____.
(A) metastasis
(B) parthenogenesis
(C) interkinesis
(D) metamorphosis

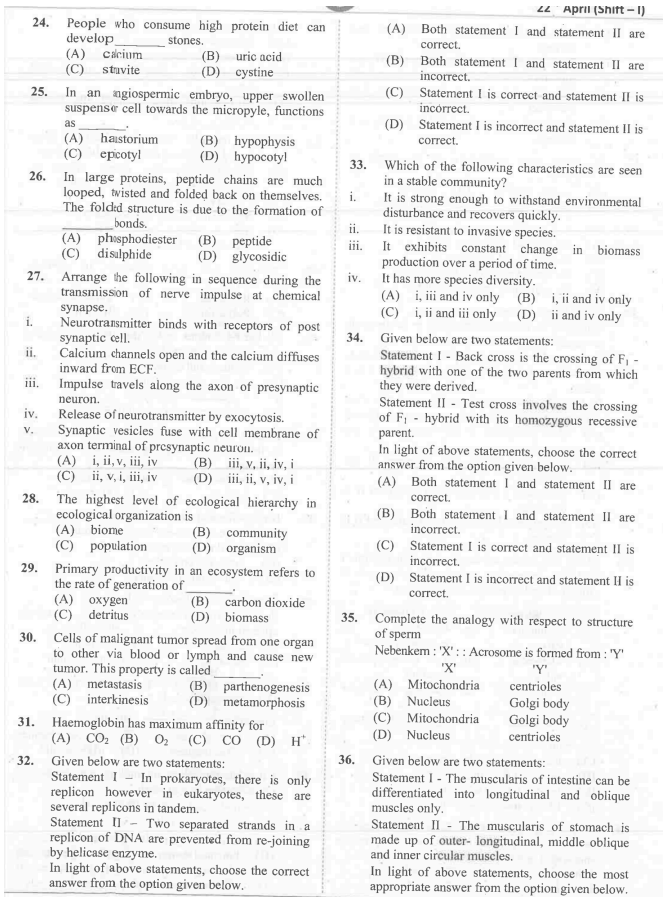
Bio_q031 key **C**

(C) interkinesis (D) metamorphosis

31. Haemoglobin has maximum affinity for
(A) CO_2 (B) O_2 (C) CO (D) H^+

Haemoglobin has maximum affinity for
(A) CO_2
(B) O_2
(C) CO
(D) H^+

Bio_q032 key **C**



22 April (Shift – I)

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(A) calcium (B) uric acid
(C) struvite (D) cystine

25. In an angiospermic embryo, upper swollen suspensor cell towards the micropyle, functions as _____.
(A) haustorium (B) hypophysis
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26. In large proteins, peptide chains are much looped, twisted and folded back on themselves. The folded structure is due to the formation of _____bonds.
(A) phosphodiester (B) peptide
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iii. Impulse travels along the axon of presynaptic neuron.
iv. Release of neurotransmitter by exocytosis.
v. Synaptic vesicles fuse with cell membrane of axon terminal of presynaptic neuron.
(A) i, ii, v, iii, iv (B) iii, v, ii, iv, i
(C) ii, v, i, iii, iv (D) iii, ii, v, iv, i

28. The highest level of ecological hierarchy in ecological organization is
(A) biome (B) community
(C) population (D) organism

29. Primary productivity in an ecosystem refers to the rate of generation of _____.
(A) oxygen (B) carbon dioxide
(C) detritus (D) biomass

30. Cells of malignant tumor spread from one organ to other via blood or lymph and cause new tumor. This property is called _____.
(A) metastasis (B) parthenogenesis
(C) interkinesis (D) metamorphosis

31. Haemoglobin has maximum affinity for
(A) CO_2 (B) O_2 (C) CO (D) H^+

32. Given below are two statements:
Statement I – In prokaryotes, there is only replicon however in eukaryotes, these are several replicons in tandem.
Statement II – Two separated strands in a replicon of DNA are prevented from re-joining by helicase enzyme.
In light of above statements, choose the correct answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

33. Which of the following characteristics are seen in a stable community?
i. It is strong enough to withstand environmental disturbance and recovers quickly.
ii. It is resistant to invasive species.
iii. It exhibits constant change in biomass production over a period of time.
iv. It has more species diversity.
(A) i, iii and iv only (B) i, ii and iv only
(C) i, ii and iii only (D) ii and iv only

34. Given below are two statements:
Statement I - Back cross is the crossing of F_1 - hybrid with one of the two parents from which they were derived.
Statement II - Test cross involves the crossing of F_1 - hybrid with its homozygous recessive parent.
In light of above statements, choose the correct answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

35. Complete the analogy with respect to structure of sperm
Nebenkern : 'X' : : Acrosome is formed from : 'Y'
'X' 'Y'
(A) Mitochondria centrioles
(B) Nucleus Golgi body
(C) Mitochondria Golgi body
(D) Nucleus centrioles

36. Given below are two statements:
Statement I - The muscularis of intestine can be differentiated into longitudinal and oblique muscles only.
Statement II - The muscularis of stomach is made up of outer- longitudinal, middle oblique and inner circular muscles.
In light of above statements, choose the most appropriate answer from the option given below.

Given below are two statements:

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In light of above statements, choose the correct answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q033 key **B**

33. Which of the following characteristics are seen in a stable community?

i. It is strong enough to withstand environmental disturbance and recovers quickly.
ii. It is resistant to invasive species.
iii. It exhibits constant change in biomass production over a period of time.
iv. It has more species diversity.

(A) i, iii and iv only (B) i, ii and iv only
(C) i, ii and iii only (D) ii and iv only

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i. It is strong enough to withstand environmental disturbance and recovers quickly.
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iii. It exhibits constant change in biomass production over a period of time.
iv. It has more species diversity.
(A) i, iii and iv only
(B) i, ii and iv only
(C) i, ii and iii only
(D) ii and iv only

Bio_q034 key **A**

34. Given below are two statements:
Statement I - Back cross is the crossing of F_1 - hybrid with one of the two parents from which they were derived.
Statement II - Test cross involves the crossing of F_1 - hybrid with its homozygous recessive parent.
In light of above statements, choose the correct answer from the option given below.

(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Given below are two statements:

Statement I - Back cross is the crossing of F_1 - hybrid with one of the two parents from which they were derived.

Statement II - Test cross involves the crossing of F_1 - hybrid with its homozygous recessive parent.

In light of above statements, choose the correct answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q035 key **C**

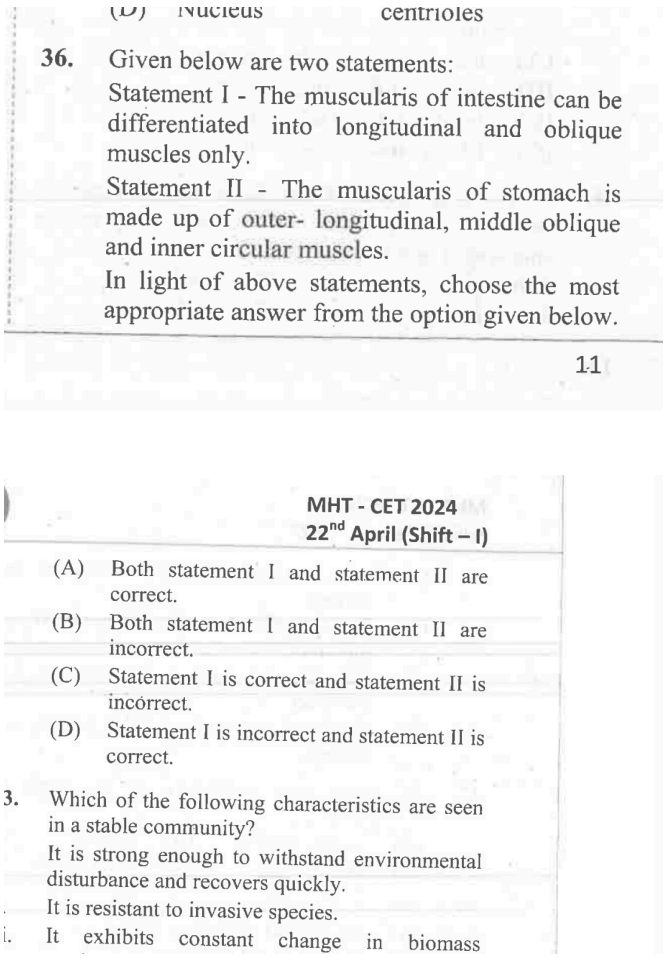
35. Complete the analogy with respect to structure of sperm
Nebenkern : 'X' : : Acrosome is formed from : 'Y'

	'X'	'Y'
(A)	Mitochondria	centrioles
(B)	Nucleus	Golgi body
(C)	Mitochondria	Golgi body
(D)	Nucleus	centrioles

Complete the analogy with respect to structure of sperm

Nebenkern: 'X':: Acrosome is formed from: 'Y'
(A) Mitochondria centrioles
(B) Nucleus Golgi body
(C) Mitochondria Golgi body
(D) Nucleus centrioles

Bio_q036 key **B** OPTION_EMPTY



(D) Nucleus centrioles

36. Given below are two statements:
Statement I - The muscularis of intestine can be differentiated into longitudinal and oblique muscles only.
Statement II - The muscularis of stomach is made up of outer- longitudinal, middle oblique and inner circular muscles.
In light of above statements, choose the most appropriate answer from the option given below.

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(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

3. Which of the following characteristics are seen in a stable community?
It is strong enough to withstand environmental disturbance and recovers quickly.
It is resistant to invasive species.
It exhibits constant change in biomass

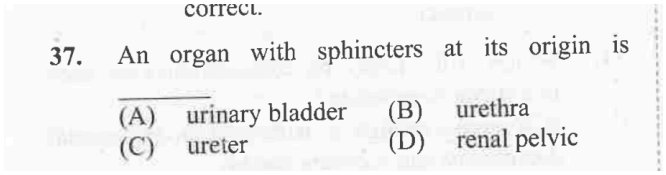
Given below are two statements:

Statement I - The muscularis of intestine can be differentiated into longitudinal and oblique muscles only.

Statement II - The muscularis of stomach is made up of outer- longitudinal, middle oblique and inner circular muscles.

In light of above statements, choose the most appropriate answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q037 key **B**

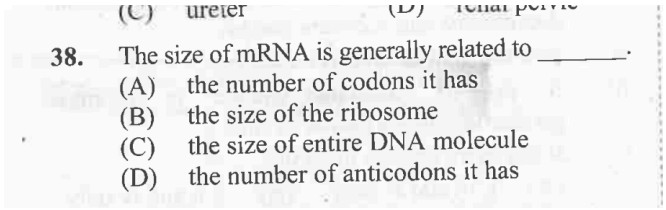


correct.

37. An organ with sphincters at its origin is _____
(A) urinary bladder (B) urethra
(C) ureter (D) renal pelvic

An organ with sphincters at its origin is
(A) urinary bladder
(B) urethra
(C) ureter
(D) renal pelvic

Bio_q038 key **A**



(C) ureter (D) renal pelvic

38. The size of mRNA is generally related to _____.
(A) the number of codons it has
(B) the size of the ribosome
(C) the size of entire DNA molecule
(D) the number of anticodons it has

The size of mRNA is generally related to _____.
(A) the number of codons it has
(B) the size of the ribosome
(C) the size of entire DNA molecule
(D) the number of anticodons it has

Bio_q039 key **A**

(D) the number of anticodons it has

39. Given below are two statements:
Statement I - Reactions involved in Krebs cycle are anabolic and catabolic.
Statement II - During oxidation of acetyl Co-A, stepwise oxidation of acetyl part of acetyl Co-A occurs.
In light of above statements, choose the most appropriate answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is

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(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q040 key **A**

incorrect.
(D) Statement I is incorrect and statement II is correct.

40. Match the valves of human heart in column I with the respective opening they guard in column II and select the correct option.

	Column I		Column II
i.	Eustachian valve	a.	Opening of pulmonary aorta
ii.	Thebesian valve	b.	Left atrioventricular aperture
iii.	Mitral valve	c.	opening of inferior vena cava
iv.	Semilunar valve	d.	opening of coronary sinus

(A) i-c ii-d iii-b iv-a
(B) i-c ii-d iii-a iv-b
(C) i-d ii-c iii-b iv-a
(D) i-b ii-a iii-c iv-d

Match the valves of human heart in column I with the respective opening they guard in column II and select the correct option.
Column I:
i. Eustachian valve |
ii. Thebesian valve |
iii. Mitral valve |
iv. Semilunar valve
Column II: a. Opening of pulmonary aorta | b. Left atrioventricular aperture | c. opening of inferior vena cava | d. opening of coronary sinus
(A) i-c ii-d iii-b iv-a
(B) i-c ii-d iii-a iv-b
(C) i-d ii-c iii-b iv-a
(D) i-b ii-a iii-c iv-d

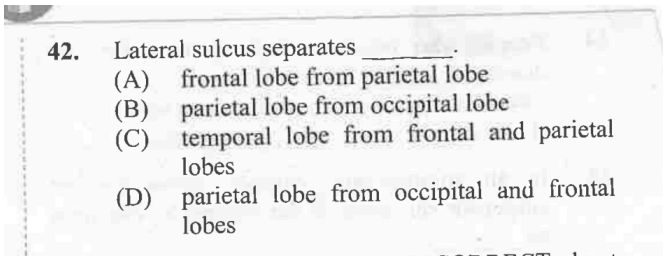
Bio_q041 key **A**

(D) i-b ii-a iii-c iv-d

41. The _____ gene from *Bacillus thuringiensis* produces a protein that forms crystalline inclusions in bacterial spores.
(A) *cry* (B) *myc*
(C) *ras* (D) *nif*

The _____ gene from Bacillus thuringiensis produces a protein that forms crystalline inclusions in bacterial spores.
(A) cry
(B) myc
(C) ras
(D) nif

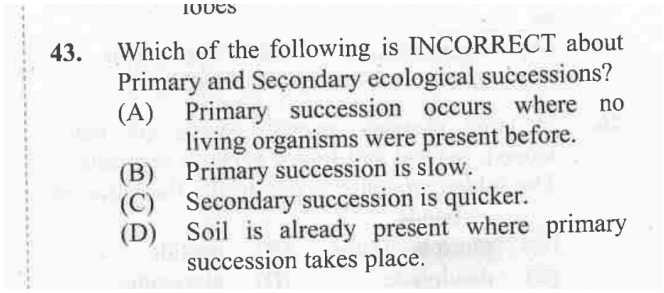
Bio_q042 key **C**



42. Lateral sulcus separates _____.
(A) frontal lobe from parietal lobe
(B) parietal lobe from occipital lobe
(C) temporal lobe from frontal and parietal lobes
(D) parietal lobe from occipital and frontal lobes

Lateral sulcus separates _____.
(A) frontal lobe from parietal lobe
(B) parietal lobe from occipital lobe
(C) temporal lobe from frontal and parietal lobes
(D) parietal lobe from occipital and frontal lobes

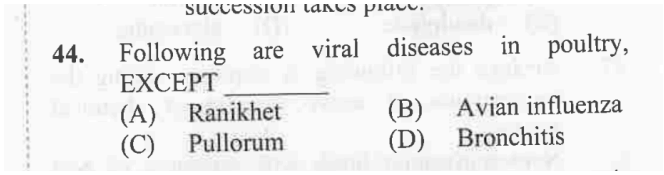
Bio_q043 key **D**



43. Which of the following is INCORRECT about Primary and Secondary ecological successions?
(A) Primary succession occurs where no living organisms were present before.
(B) Primary succession is slow.
(C) Secondary succession is quicker.
(D) Soil is already present where primary succession takes place.

Which of the following is INCORRECT about Primary and Secondary ecological successions?
(A) Primary succession occurs where no living organisms were present before.
(B) Primary succession is slow.
(C) Secondary succession is quicker.
(D) Soil is already present where primary succession takes place.

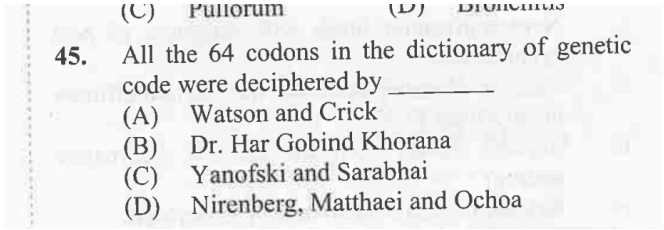
Bio_q044 key **C**



44. Following are viral diseases in poultry, EXCEPT _____
(A) Ranikhet (B) Avian influenza
(C) Pullorum (D) Bronchitis

Following are viral diseases in poultry, EXCEPT _____.
(A) Ranikhet
(B) Avian influenza
(C) Pullorum
(D) Bronchitis

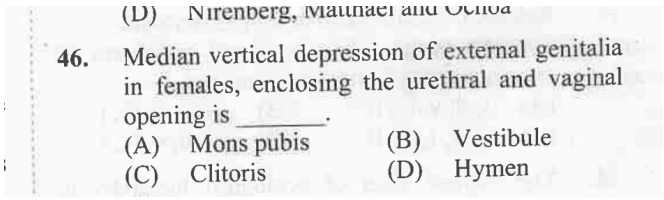
Bio_q045 key **D** SOLVER_DISAGREES_WITH_BOOK



45. All the 64 codons in the dictionary of genetic code were deciphered by _____
(A) Watson and Crick
(B) Dr. Har Gobind Khorana
(C) Yanofski and Sarabhai
(D) Nirenberg, Matthaei and Ochoa

All the 64 codons in the dictionary of genetic code were deciphered by _____.
(A) Watson and Crick
(B) Dr. Har Gobind Khorana
(C) Yanofski and Sarabhai
(D) Nirenberg, Matthaei and Ochoa

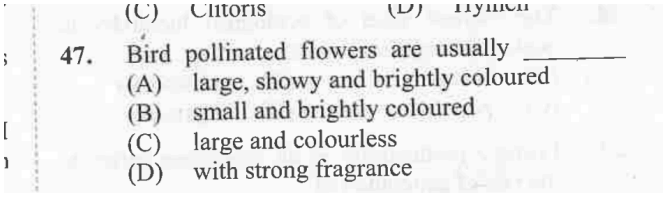
Bio_q046 key **B**



46. Median vertical depression of external genitalia in females, enclosing the urethral and vaginal opening is _____.
(A) Mons pubis (B) Vestibule
(C) Clitoris (D) Hymen

Median vertical depression of external genitalia in females, enclosing the urethral and vaginal opening is _____.
(A) Mons pubis
(B) Vestibule
(C) Clitoris
(D) Hymen

Bio_q047 key **A**



47. Bird pollinated flowers are usually _____
(A) large, showy and brightly coloured
(B) small and brightly coloured
(C) large and colourless
(D) with strong fragrance

Bird pollinated flowers are usually _____.
(A) large, showy and brightly coloured
(B) small and brightly coloured
(C) large and colourless
(D) with strong fragrance

Bio_q048 key **B**

48. Complete the following analogy.
Parietal wall of Bowman's capsule : squamous epithelium : : PCT : _____ epithelium.
(A) ciliated (B) cuboidal
(C) squamous (D) transitional

Complete the following analogy.
Parietal wall of Bowman's capsule: squamous epithelium :: PCT: _____ epithelium.
(A) ciliated
(B) cuboidal
(C) squamous
(D) transitional

Bio_q049 key **A**

49. Complete the following reaction by replacing (Y) with proper word.
Emulsified fats $\xrightarrow{\text{Lipases}}$ Fatty acids + (Y)
(A) monoglycerides (B) monosaccharides
(C) disaccharides (D) triglycerides

Complete the following reaction by replacing (Y) with proper word.
Emulsified fats $\xrightarrow{\text{Lipases}}$ Fatty acids + (Y)
(A) monoglycerides
(B) monosaccharides
(C) disaccharides
(D) triglycerides

Bio_q050 key **B**

50. Complete the following analogy with respect to classification of mammals.
Marsupials : (i) :: monotremes : (ii)

	(i)	(ii)
(A)	pouched mammals	placental mammals
(B)	pouched mammals	egg laying mammals
(C)	placental mammals	egg laying mammals
(D)	placental mammals	pouched mammals

Complete the following analogy with respect to classification of mammals.
Marsupials: (i) :: monotremes: (ii)
(i) (ii)
(A) pouched mammals placental mammals
(B) pouched mammals egg laying mammals
(C) placental mammals egg laying mammals
(D) placental mammals pouched mammals

Bio_q051 key **B**

51. While studying flowering behaviour in _____ and _____ plant, Garner and Allard discovered photoperiodism.
(A) rice, wheat
(B) tobacco, soyabean
(C) tomato, soyabean
(D) wheat, tobacco

While studying flowering behaviour in _____ and _____ plant, Garner and Allard discovered photoperiodism.
(A) rice, wheat
(B) tobacco, soyabean
(C) tomato, soyabean
(D) wheat, tobacco

Bio_q052 key **D** V9_NEEDS_HUMAN

Bacillus, Pseudomonas, Micrococcus, Aspergillus, Agrobacterium and Klebsiella are all listed as phosphate solubilisers in the textbook; key D (Klebsiella and Beijerinckia) depends on the textbook list, which I cannot verify from the crop.

52. Which of the following bacteria do not solubilize the insoluble rock phosphate?
(A) *Bacillus polymyxa* and *Pseudomonas striata*.
(B) *Micrococcus* and *Aspergillus* spp.
(C) *Pseudomonas striata* and *Agrobacterium*.
(D) *Klebsiella* and *Beijerinckia*.

Which of the following bacteria do not solubilize the insoluble rock phosphate?
(A) Bacillus polymyxa and Pseudomonas striata.
(B) Micrococcus and Aspergillus spp.
(C) Pseudomonas striata and Agrobacterium.
(D) Klebsiella and Beijerinckia.

Bio_q053 key **A** SOLVER_DISAGREES_WITH_BOOK, V9_NEEDS_HUMAN

Key A (PDGF) promotes plaque growth, so it looks wrong, but tPA treats clots rather than atherosclerosis and no option is clearly the textbook answer.

(D) *Klebsiella* and *Beijerinckia*.

53. Atherosclerosis can be treated with recombinant protein called _____
(A) platelet derived growth factor.
(B) interleukin -1 receptor.
(C) tissue plasminogen activator.
(D) macrophage activating factor.

Atherosclerosis can be treated with recombinant protein called
(A) platelet derived growth factor.
(B) interleukin-1 receptor.
(C) tissue plasminogen activator.
(D) macrophage activating factor.

Bio_q054 key **B**

(D) macrophage activating factor.

54. Match elements given in Column I with their deficiency symptoms shown by plants given in Column II

	Column I		Column II
i.	Chlorine	a.	Brown heart disease
ii.	Boron	b.	Malformed leaves
iii.	Zinc	c.	Poor growth of the plant
iv.	Copper	d.	Die-back of shoots

(A) i-c ii-a iii-d iv-b
(B) i-c ii-a iii-b iv-d
(C) i-c ii-b iii-d iv-a
(D) i-b ii-a iii-d iv-c

Match elements given in Column I with their deficiency symptoms shown by plants given in Column II
Column I:
i. Chlorine |
ii. Boron |
iii. Zinc |
iv. Copper
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(A) i-c ii-a iii-d iv-b
(B) i-c ii-a iii-b iv-d
(C) i-c ii-b iii-d iv-a
(D) i-b ii-a iii-d iv-c

Bio_q055 key **A** SOLVER_DISAGREES_WITH_BOOK, V9_NEEDS_HUMAN

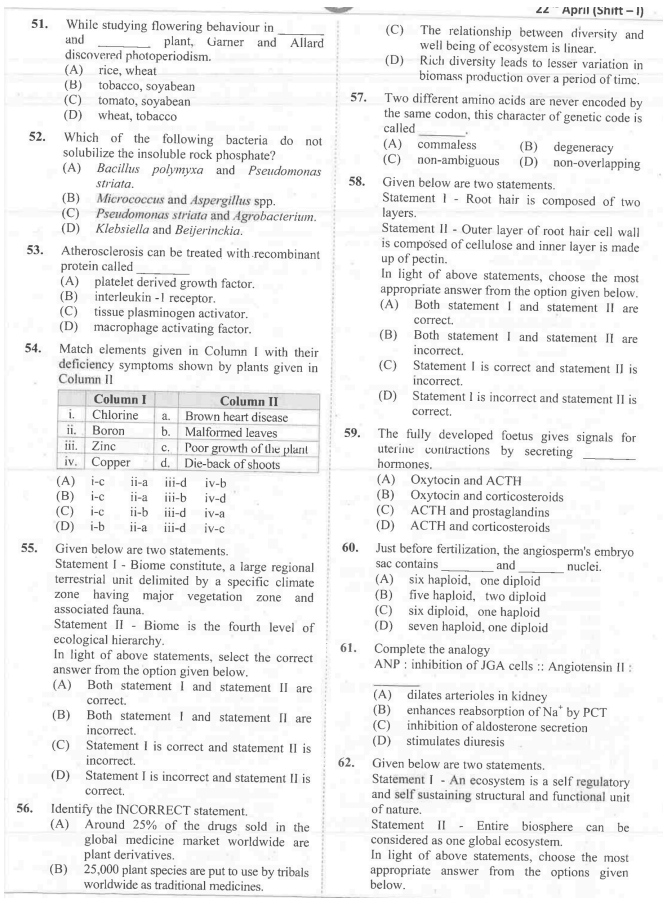
Statement I is correct, but biome is the fifth level in the usual hierarchy (ecosystem comes before it), so statement II is doubtful and key A rests on one textbook's list.

(D) i-b ii-a iii-d iv-c

55. Given below are two statements.
Statement I - Biome constitute, a large regional terrestrial unit delimited by a specific climate zone having major vegetation zone and associated fauna.
Statement II - Biome is the fourth level of ecological hierarchy.
In light of above statements, select the correct answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Given below are two statements. Statement I - Biome constitute, a large regional terrestrial unit delimited by a specific climate zone having major vegetation zone and associated fauna. Statement II - Biome is the fourth level of ecological hierarchy. In light of above statements, select the correct answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q056 key **C**



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51. While studying flowering behaviour in _____ and _____ plant, Garner and Allard discovered photoperiodism.
(A) rice, wheat
(B) tobacco, soyabean
(C) tomato, soyabean
(D) wheat, tobacco

52. Which of the following bacteria do not solubilize the insoluble rock phosphate?
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(D) *Klebsiella* and *Beijerinckia*.

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54. Match elements given in Column I with their deficiency symptoms shown by plants given in Column II

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iii.	Zinc	c.	Poor growth of the plant
iv.	Copper	d.	Die-back of shoots

(A) i-c ii-a iii-d iv-b
(B) i-c ii-a iii-b iv-d
(C) i-c ii-b iii-d iv-a
(D) i-b ii-a iii-d iv-c

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(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

56. Identify the INCORRECT statement.
(A) Around 25% of the drugs sold in the global medicine market worldwide are plant derivatives.
(B) 25,000 plant species are put to use by tribals worldwide as traditional medicines.
(C) The relationship between diversity and well being of ecosystem is linear.
(D) Rich diversity leads to lesser variation in biomass production over a period of time.

57. Two different amino acids are never encoded by the same codon, this character of genetic code is called _____.
(A) commaless (B) degeneracy
(C) non-ambiguous (D) non-overlapping

58. Given below are two statements.
Statement I - Root hair is composed of two layers.
Statement II - Outer layer of root hair cell wall is composed of cellulose and inner layer is made up of pectin.
In light of above statements, choose the most appropriate answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

59. The fully developed foetus gives signals for uterine contractions by secreting _____ hormones.
(A) Oxytocin and ACTH
(B) Oxytocin and corticosteroids
(C) ACTH and prostaglandins
(D) ACTH and corticosteroids

60. Just before fertilization, the angiosperm's embryo sac contains _____ and _____ nuclei.
(A) six haploid, one diploid
(B) five haploid, two diploid
(C) six diploid, one haploid
(D) seven haploid, one diploid

61. Complete the analogy
ANP : inhibition of JGA cells :: Angiotensin II : _____
(A) dilates arterioles in kidney
(B) enhances reabsorption of Na^+ by PCT
(C) inhibition of aldosterone secretion
(D) stimulates diuresis

62. Given below are two statements.
Statement I - An ecosystem is a self regulatory and self sustaining structural and functional unit of nature.
Statement II - Entire biosphere can be considered as one global ecosystem.
In light of above statements, choose the most appropriate answer from the options given below.

Identify the INCORRECT statement.

(A) Around 25% of the drugs sold in the global medicine market worldwide are plant derivatives.

(B) 25,000 plant species are put to use by tribals worldwide as traditional medicines.

(C) The relationship between diversity and well being of ecosystem is linear.

(D) Rich diversity leads to lesser variation in biomass production over a period of time.

Bio_q057 key **C**

biomass production over a period of time.

57. Two different amino acids are never encoded by the same codon, this character of genetic code is called _____.
(A) commaless (B) degeneracy
(C) non-ambiguous (D) non-overlapping

Two different amino acids are never encoded by the same codon, this character of genetic code is called

(A) commaless

(B) degeneracy

(C) non-ambiguous

(D) non-overlapping

Bio_q058 key **C**

(C) non-ambiguous (D) non-overlapping

58. Given below are two statements.
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In light of above statements, choose the most appropriate answer from the option given below.
(A) Both statement I and statement II are correct.
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(C) Statement I is correct and statement II is incorrect.
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(A) Both statement I and statement II are correct.

(B) Both statement I and statement II are incorrect.

(C) Statement I is correct and statement II is incorrect.

(D) Statement I is incorrect and statement II is correct.

Bio_q059 key **D**

correct.

59. The fully developed foetus gives signals for uterine contractions by secreting _____ hormones.
(A) Oxytocin and ACTH
(B) Oxytocin and corticosteroids
(C) ACTH and prostaglandins
(D) ACTH and corticosteroids

The fully developed foetus gives signals for uterine contractions by secreting hormones.
(A) Oxytocin and ACTH
(B) Oxytocin and corticosteroids
(C) ACTH and prostaglandins
(D) ACTH and corticosteroids

Bio_q060 key **A**

(D) ACTH and corticosteroids

60. Just before fertilization, the angiosperm's embryo sac contains _____ and _____ nuclei.
(A) six haploid, one diploid
(B) five haploid, two diploid
(C) six diploid, one haploid
(D) seven haploid, one diploid

Just before fertilization, the angiosperm's embryo sac contains _____ and _____ nuclei.
(A) six haploid, one diploid
(B) five haploid, two diploid
(C) six diploid, one haploid
(D) seven haploid, one diploid

Bio_q061 key **B**

(D) seven haploid, one diploid

61. Complete the analogy
ANP : inhibition of JGA cells :: Angiotensin II : _____
(A) dilates arterioles in kidney
(B) enhances reabsorption of Na^+ by PCT
(C) inhibition of aldosterone secretion
(D) stimulates diuresis

Complete the analogy
ANP: inhibition of JGA cells :: Angiotensin II:
(A) dilates arterioles in kidney
(B) enhances reabsorption of Na^+ by PCT
(C) inhibition of aldosterone secretion
(D) stimulates diuresis

Bio_q062 key **A** OPTION_EMPTY, V9_UNSOLVABLE

Both statements are true (A if the options are the standard set), but the options are not visible in crop or crop_continued, so the stored option text cannot be verified.

(D) stimulates diuresis

62. Given below are two statements.
Statement I - An ecosystem is a self regulatory and self sustaining structural and functional unit of nature.
Statement II - Entire biosphere can be considered as one global ecosystem.
In light of above statements, choose the most appropriate answer from the options given below.

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(C) The relationship between diversity and well being of ecosystem is linear.
(D) Rich diversity leads to lesser variation in biomass production over a period of time.

7. Two different amino acids are never encoded by the same codon, this character of genetic code is called _____.
(A) commaless (B) degeneracy
(C) non-ambiguous (D) non-overlapping

3. Given below are two statements.
Statement I - Root hair is composed of two layers.
Statement II - Outer layer of root hair cell wall is composed of cellulose and inner layer is made

Given below are two statements. Statement I - An ecosystem is a self regulatory and self sustaining structural and functional unit of nature. Statement II - Entire biosphere can be considered as one global ecosystem. In light of above statements, choose the most appropriate answer from the options given below.

(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q063 key **C**

correct.

63. The number of reduced coenzymes $NADH+H^+$ formed during complete oxidation of one molecule of glucose.
(A) 2 (B) 8 (C) 10 (D) 12

The number of reduced coenzymes $NADH + H^+$ formed during complete oxidation of one molecule of glucose.

(A) 2
(B) 8
(C) 10
(D) 12

Bio_q064 key **C**

(A) 2 (B) 8 (C) 10 (D) 12

64. Monohybrid crosses are useful in demonstrating which of the following laws of inheritance suggested by Mendel?
(A) Law of dominance and law of independent assortment.
(B) Law of segregation and law of independent assortment.
(C) Law of purity of gamete and law of dominance.
(D) Law of independent assortment only.

Monohybrid crosses are useful in demonstrating which of the following laws of inheritance suggested by Mendel?

(A) Law of dominance and law of independent assortment.
(B) Law of segregation and law of independent assortment.
(C) Law of purity of gamete and law of dominance.
(D) Law of independent assortment only.

Bio_q065 key **B**

(D) Law of independent assortment only.

65. Enzymes are needed only in small quantities to catalyse reactions because
(A) they act very fast.
(B) they remain unchanged at the end of the reaction.
(C) they are made up to proteins.
(D) they are temperature and pH sensitive.

Enzymes are needed only in small quantities to catalyse reactions because
(A) they act very fast.
(B) they remain unchanged at the end of the reaction.
(C) they are made up to proteins.
(D) they are temperature and pH sensitive.

Bio_q066 key **B** V9_NEEDS_HUMAN

The question is ambiguous (the amide nitrogen is not an amino group in the strict sense), so key B (two) depends on which textbook convention is intended.

(D) they are temperature and pH sensitive.

66. The number of amino groups in the amino acids present in amides is _____.
(A) one (B) two
(C) three (D) four

The number of amino groups in the amino acids present in amides is _____.
(A) one
(B) two
(C) three
(D) four

Bio_q067 key **D**

(C) three (D) four

67. Which of the following are the initial and final steps during the process of ecological succession?
(A) Nudation and Aggregation
(B) Invasion and Competition
(C) Competition and co-action
(D) Nudation and Stabilization

Which of the following are the initial and final steps during the process of ecological succession?
(A) Nudation and Aggregation
(B) Invasion and Competition
(C) Competition and co-action
(D) Nudation and Stabilization

Bio_q068 key **C**

(D) Nudation and Stabilization

68. Match column I with column II and select the correct option

	Column I		Column II
i.	Carrot and spinach	a.	High protein content
ii.	Bitter gourd	b.	More Vit A and minerals
iii.	Maize	c.	More Vit C
iv.	Wheat Atlas-66	d.	Twice the amount of lysine and tryptophan

(A) i-d ii-a iii-b iv-c
(B) i-d ii-a iii-c iv-b
(C) i-b ii-c iii-d iv-a
(D) i-b ii-c iii-a iv-d

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Match column I with column II and select the correct option Column I:
i. Carrot and spinach |
ii. Bitter gourd |
iii. Maize |
iv. Wheat Atlas-66 | Column II: a. High protein content | b. More Vit A and minerals | c. More Vit C | d. Twice the amount of lysine and tryptophan
(A) i-d ii-a iii-b iv-c
(B) i-d ii-a iii-c iv-b
(C) i-b ii-c iii-d iv-a
(D) i-b ii-c iii-a iv-d

Bio_q069 key **A**

69. Hypersecretion of glucocorticoid leads to _____.
(A) Cushing's disease
(B) Addison's disease
(C) Grave's disease
(D) Bleeders' disease

Hypersecretion of glucocorticoid leads to _____.
(A) Cushing's disease
(B) Addison's disease
(C) Grave's disease
(D) Bleeders' disease

Bio_q070 key **B**

70. The restriction enzymes used as molecular scissors are type of _____
(A) ligases (B) nucleases
(C) protease (D) carboxylases

The restriction enzymes used as molecular scissors are type of _____
(A) ligases
(B) nucleases
(C) protease
(D) carboxylases

Bio_q071 key **A**

71. Given below are two statements.
Statement I – *Calotropis* growing in abandoned fields is never consumed by cattle and goats.
Statement II – *Calotropis* produces highly poisonous cardiac glycosides.
In light of above statements, select the most appropriate answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Given below are two statements. Statement I - Calotropis growing in abandoned fields is never consumed by cattle and goats. Statement II - Calotropis produces highly poisonous cardiac glycosides. In light of above statements, select the most appropriate answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q072 key **A**

72. Phenotypically tall plants can be obtained from genetically dwarf maize plants by application of _____.
(A) Gibberellins (B) Auxins
(C) Ethylene (D) ABA

Phenotypically tall plants can be obtained from genetically dwarf maize plants by application of _____.
(A) Gibberellins
(B) Auxins
(C) Ethylene
(D) ABA

Bio_q073 key **A**

73. Islets of Langerhans are _____.
(A) groups of endocrine cells of pancreas
(B) group of cells in the stroma of thyroid gland
(C) star shaped glial cells of CNS
(D) grey masses within the white matter of nervous system

Islets of Langerhans are _____.
(A) groups of endocrine cells of pancreas
(B) group of cells in the stroma of thyroid gland
(C) star shaped glial cells of CNS
(D) grey masses within the white matter of nervous system

Bio_q074 key **D**

nervous system

74. *Homo sapiens* are most closely related to _____.
(A) Lemurs (B) Tarsiers
(C) Baboons (D) Orangutan

Homo sapiens are most closely related to _____.
(A) Lemurs
(B) Tarsiers
(C) Baboons
(D) Orangutan

Bio_q075 key **C**

(C) Baboons (D) Orangutan

75. Complete the analogy with respect to reproductive system.
Male : Penis : : Female : _____.
(A) Vagina (B) Vulva
(C) Clitoris (D) Cervix

Complete the analogy with respect to reproductive system. Male: Penis :: Female: _____
(A) Vagina
(B) Vulva
(C) Clitoris
(D) Cervix

Bio_q076 key **A**

(C) Clitoris (D) Cervix

76. In a male gametophyte, ploidy of tube cell and a male gamete is _____.
(A) both are haploid
(B) both are diploid
(C) tube cell is haploid and male gamete is diploid
(D) tube cell is diploid and male gamete is haploid

In a male gametophyte, ploidy of tube cell and a male gamete is _____.
(A) both are haploid
(B) both are diploid
(C) tube cell is haploid and male gamete is diploid
(D) tube cell is diploid and male gamete is haploid

Bio_q077 key **C**

77. In which of the following, bilirubin is excreted in urine?
(A) Marasmus (B) Kwashiorkor
(C) Jaundice (D) Indigestion

In which of the following, bilirubin is excreted in urine?
(A) Marasmus
(B) Kwashiorkor
(C) Jaundice
(D) Indigestion

Bio_q078 key **B**

(C) Jaundice (D) Indigestion

78. The only 5C intermediate formed during reactions of TCA cycle is _____.
(A) Oxalosuccinic acid
(B) α-ketoglutarate
(C) Succinyl CoA
(D) Cis - Aconitate

The only 5C intermediate formed during reactions of TCA cycle is _____.
(A) Oxalosuccinic acid
(B) α -ketoglutarate
(C) Succinyl CoA
(D) Cis - Aconitate

Bio_q079 key **A**

(D) Cis - Aconitate

79. Secondary metabolites like glucosinolates are produced by cabbage to
(A) protect it from many pests
(B) attracts insects for pollination
(C) kills weeds around it
(D) improve soil fertilization

Secondary metabolites like glucosinolates are produced by cabbage to
(A) protect it from many pests
(B) attracts insects for pollination
(C) kills weeds around it
(D) improve soil fertilization

Bio_q080 key **B**

(D) improve soil fertilization

80. Tubular fluid tends to become acidic in the _____ of nephron.
(A) PCT
(B) DCT
(C) neck
(D) Henle's loop

Tubular fluid tends to become acidic in the _____ of nephron.
(A) PCT
(B) DCT
(C) neck
(D) Henle's loop

Bio_q081 key **B**

(D) Henle's loop

81. All are events of pollen-pistil interaction EXCEPT
(A) absorption of water and nutrients from the stigmatic surface.
(B) first division of pollen grain.
(C) deposition of pollen grain on stigma.
(D) germination of pollen grains to produce pollen tube.

All are events of pollen-pistil interaction EXCEPT
(A) absorption of water and nutrients from the stigmatic surface.
(B) first division of pollen grain.
(C) deposition of pollen grain on stigma.
(D) germination of pollen grains to produce pollen tube.

Bio_q082 key **B**

pollen tube.

82. Which one of the following cells secrete heparin, histamine and serotonin?
(A) Lymphocytes (B) Basophils
(C) Eosinophils (D) Thrombocytes

Which one of the following cells secrete heparin, histamine and serotonin?
(A) Lymphocytes
(B) Basophils
(C) Eosinophils
(D) Thrombocytes

Bio_q083 key **B**

(C) Eosinophils (D) Thrombocytes

83. In an antibody, disulphide bonds are present _____.
i. between two heavy chains
ii. between two light chains
iii. between the constant region of light chain and the constant region of a heavy chain
iv. between antigen binding sites of both the heavy and light chains
v. between the variable region of light chain and that of heavy chain
Select the correct answer from the options given below
(A) i and ii only (B) i and iii only
(C) ii and v only (D) ii and iv only

In an antibody, disulphide bonds are present _____.
i. between two heavy chains
ii. between two light chains
iii. between the constant region of light chain and the constant region of a heavy chain
iv. between antigen binding sites of both the heavy and light chains
v. between the variable region of light chain and that of heavy chain Select the correct answer from the options given below
(A) i and ii only
(B) i and iii only
(C) ii and v only
(D) ii and iv only

Bio_q084 key **A**

(C) ii and v only (D) ii and iv only

84. The water which percolates deep in the soil is _____.
(A) Gravitational water
(B) Capillary water
(C) Combined water
(D) Hygroscopic water

The water which percolates deep in the soil is _____.
(A) Gravitational water
(B) Capillary water
(C) Combined water
(D) Hygroscopic water

Bio_q085 key **D**

22nd April (Shift – I)

85. Select INCORRECT statements with respects transport of CO_2 by RBCs and plasma.
i. 23% of CO_2 released from tissue cells diffuse into plasma first and then into RBCs.
ii. Within RBCs carbonic acid immediately dissociates into H^+ and HCO_3^-.
iii. In RBCs CO_2 combines with H_2O to form carbonic acid.
iv. H^+ ions move out of RBCs and combine with Na^+ to form $NaHCO_3$.
v. 7% of CO_2 is transported in dissolved form as carbonic acid in plasma.
In the light of above statements choose the correct answer from option given below.
(A) i, ii and iii only (B) iii, iv and v only
(C) ii and v only (D) i and iv only

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i. 23% of CO_2 released from tissue cells diffuse into plasma first and then into RBCs.
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v. 7% of CO_2 is transported in dissolved form as carbonic acid in plasma. In the light of above statements choose the correct answer from option given below.
(A) i, ii and iii only
(B) iii, iv and v only
(C) ii and v only
(D) i and iv only

Bio_q086 key **D**

(C) ii and v only (D) i and iv

86. Dispersal of pollens is an example of
(A) gene mutation
(B) genetic drifts
(C) genetic recombination
(D) gene flow

Dispersal of pollens is an example of
(A) gene mutation
(B) genetic drifts
(C) genetic recombination
(D) gene flow

Bio_q087 key **B**

(D) gene flow

87. Given below are two statements:
Statement I - Toddy is made by fermenting fleshy pedicels of cashews nuts.
Statement II - Fenny is made by fermenting the sugar sap extracted from palm plants and coconut palm.
In light of above statements, select the correct answer from the option given below.
(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is

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(A) Both statement I and statement II are correct.
(B) Both statement I and statement II are incorrect.
(C) Statement I is correct and statement II is incorrect.
(D) Statement I is incorrect and statement II is correct.

Bio_q088 key **D**

88. Which one of the following pair of hormones is NOT antagonistic?
(A) Calcitonin and Parathormone
(B) Glucagon and Insulin
(C) Atrial Natriuretic hormone and Aldosterone
(D) Oestrogen and Progesterone

Which one of the following pair of hormones is NOT antagonistic?
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(C) Atrial Natriuretic hormone and Aldosterone
(D) Oestrogen and Progesterone

Bio_q089 key **A** V9_INVALID

Capping always comes first in hnRNA processing, so no option is correct; key A (splicing first) is contestable and the printed sequence is defective.

(D) Oestrogen and Progesterone

89. In the processing of hnRNA in eukaryotic cell, the primary transcripts are processed in the following sequence.
(A) Splicing, Capping, Tailing
(B) Tailing, Capping, Splicing
(C) Splicing, Tailing, Capping
(D) Tailing, Splicing, Capping

In the processing of hnRNA in eukaryotic cell, the primary transcripts are processed in the following sequence.
(A) Splicing, Capping, Tailing
(B) Tailing, Capping, Splicing
(C) Splicing, Tailing, Capping
(D) Tailing, Splicing, Capping

Bio_q090 key **D**

(D) Tailing, Splicing, Capping

90. Hormone causing vigorous contraction of myometrium to initiate parturition is _____.
(A) hCG (B) ACTH
(C) corticosteroids (D) oxytocin

15

Hormone causing vigorous contraction of myometrium to initiate parturition is _____.
(A) hCG
(B) ACTH
(C) corticosteroids
(D) oxytocin

MHT - CET 2024
22nd April (Shift – I)

5. Select INCORRECT statements with respects transport of CO_2 by RBCs and plasma.
23% of CO_2 released from tissue cells diffuse into plasma first and then into RBCs.
Within RBCs carbonic acid immediately dissociates into H^+ and HCO_3^-.
i. In RBCs CO_2 combines with H_2O to form carbonic acid.
. H^+ ions move out of RBCs and combine with Na^+ to form $NaHCO_3$.
7% of CO_2 is transported in dissolved form as carbonic acid in plasma.
In the light of above statements choose the correct answer from option given below.
(A) i, ii and iii only (B) iii, iv and v only
(C) ii and v only (D) i and iv only

Bio_q091 key **D**

Solved Papers 2024

91. Which plant hormone is called antitranspirant?
(A) Gibberellins (B) Auxin
(C) Ethylene (D) ABA

Which plant hormone is called antitranspirant?
(A) Gibberellins
(B) Auxin
(C) Ethylene
(D) ABA

Bio_q092 key **D** V9_NEEDS_HUMAN

ADH lowers blood osmolarity (i) and ANP is often cited as also lowering it (iv), but no option lists i and iv, and the ANP effect on osmolarity is textbook-dependent, so D (i only) is uncertain.

(C) Ethylene (D) ABA

92. Blood osmolarity decreases with the secretion of
i. ADH
ii. aldosterone
iii. angiotensin II
iv. ANP
(A) i and ii only (B) ii and iii only
(C) ii and iv only (D) i only

Blood osmolarity decreases with the secretion of
i. ADH
ii. aldosterone
iii. angiotensin II
iv. ANP
(A) i and ii only
(B) ii and iii only
(C) ii and iv only
(D) i only

Bio_q093 key **B**

(C) ii and iv only (D) i only

93. Blood plasma transports oxygen in dissolved state, percentage of which is _____.
(A) 7% (B) 3%
(C) 23% (D) 10%

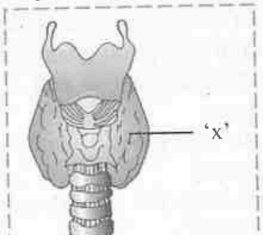
Blood plasma transports oxygen in dissolved state, percentage of which is _____.
(A) 7%
(B) 3%
(C) 23%
(D) 10%

Bio_q094 key **A**

(C) 23% (D) 10%

94. Which of the following statements are correct?
i. X chromosome has large amount of euchromatin.
ii. heterochromatin is genetically inert.
iii. Both X and Y chromosomes are homologous.
iv. Crossing over does not take place in sex chromosomes in female.
Choose the correct option.
(A) i and ii only (B) ii and iii only
(C) i, ii and iii only (D) iii and iv only

95. Identify label 'X' from the given diagram and name the cells, which regulate concentration of calcium and phosphorous in the blood and select the correct option given below.



nt?
(C) Parathyroid gland - Acidophils
(D) Parathyroid gland - chromatophores

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iii. Both X and Y chromosomes are homologous.
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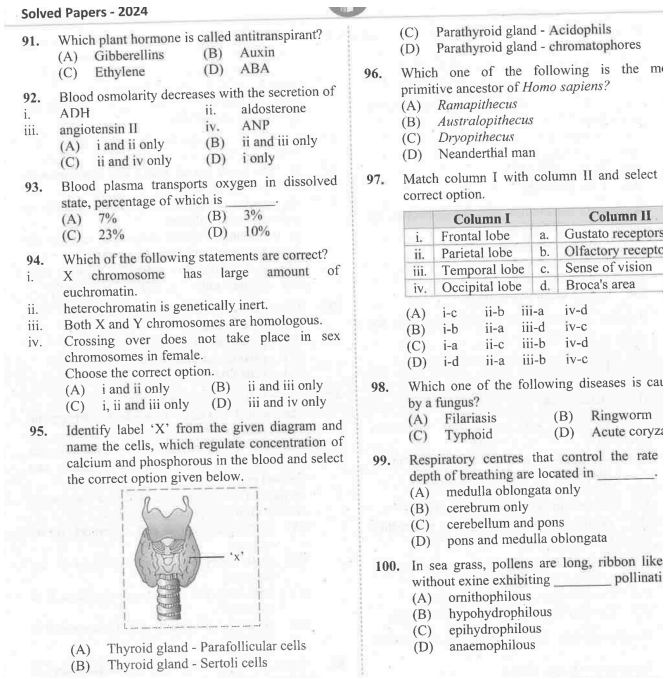
Choose the correct option.
(A) i and ii only
(B) ii and iii only
(C) i, ii and iii only
(D) iii and iv only

Bio_q095 key **A**



Solved Papers - 2024

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iii. Both X and Y chromosomes are homologous.
iv. Crossing over does not take place in sex chromosomes in female.
Choose the correct option.
(A) i and ii only (B) ii and iii only
(C) i, ii and iii only (D) iii and iv only

95. Identify label 'X' from the given diagram and name the cells, which regulate concentration of calcium and phosphorous in the blood and select the correct option given below.

'x'

(A) Thyroid gland - Parafollicular cells
(B) Thyroid gland - Sertoli cells
(C) Parathyroid gland - Acidophils
(D) Parathyroid gland - chromatophores

96. Which one of the following is the m primitive ancestor of *Homo sapiens?*
(A) *Ramapithecus*
(B) *Australopithecus*
(C) *Dryopithecus*
(D) Neanderthal man

97. Match column I with column II and select correct option.

	Column I		Column II
i.	Frontal lobe	a.	Gustato receptors
ii.	Parietal lobe	b.	Olfactory recepto
iii.	Temporal lobe	c.	Sense of vision
iv.	Occipital lobe	d.	Broca's area

(A) i-c ii-b iii-a iv-d
(B) i-b ii-a iii-d iv-c
(C) i-a ii-c iii-b iv-d
(D) i-d ii-a iii-b iv-c

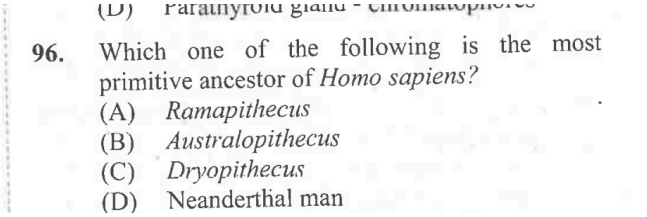
98. Which one of the following diseases is ca by a fungus?
(A) Filariasis (B) Ringworm
(C) Typhoid (D) Acute coryz

99. Respiratory centres that control the rate depth of breathing are located in _____.
(A) medulla oblongata only
(B) cerebrum only
(C) cerebellum and pons
(D) pons and medulla oblongata

100. In sea grass, pollens are long, ribbon like without exine exhibiting _____ pollinati
(A) ornithophilous
(B) hypohydrophilous
(C) epihydrophilous
(D) anaemophilous

Identify label 'X' from the given diagram and name the cells, which regulate concentration of calcium and phosphorous in the blood and select the correct option given below. [figure]
(A) Thyroid gland - Parafollicular cells
(B) Thyroid gland - Sertoli cells
(C) Parathyroid gland - Acidophils
(D) Parathyroid gland - chromatophores

Bio_q096 key **C**

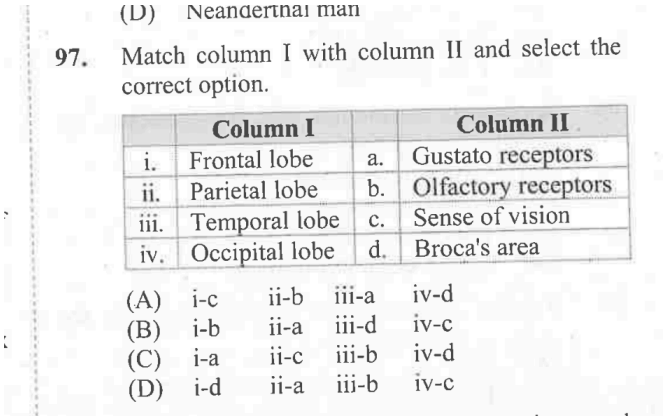


(D) Parathyroid gland - chromatophores

96. Which one of the following is the most primitive ancestor of *Homo sapiens?*
(A) *Ramapithecus*
(B) *Australopithecus*
(C) *Dryopithecus*
(D) Neanderthal man

Which one of the following is the most primitive ancestor of Homo sapiens?
(A) Ramapithecus
(B) Australopithecus
(C) Dryopithecus
(D) Neanderthal man

Bio_q097 key **D**



(D) Neanderthal man

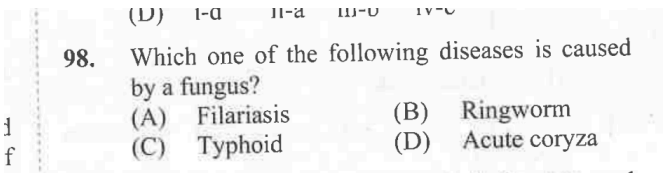
97. Match column I with column II and select the correct option.

	Column I		Column II
i.	Frontal lobe	a.	Gustato receptors
ii.	Parietal lobe	b.	Olfactory receptors
iii.	Temporal lobe	c.	Sense of vision
iv.	Occipital lobe	d.	Broca's area

(A) i-c ii-b iii-a iv-d
(B) i-b ii-a iii-d iv-c
(C) i-a ii-c iii-b iv-d
(D) i-d ii-a iii-b iv-c

Match column I with column II and select the correct option.
Column I:
i. Frontal lobe |
ii. Parietal lobe |
iii. Temporal lobe |
iv. Occipital lobe
Column II: a. Gustato receptors | b. Olfactory receptors | c. Sense of vision | d. Broca's area
(A) i-c ii-b iii-a iv-d
(B) i-b ii-a iii-d iv-c
(C) i-a ii-c iii-b iv-d
(D) i-d ii-a iii-b iv-c

Bio_q098 key **B**

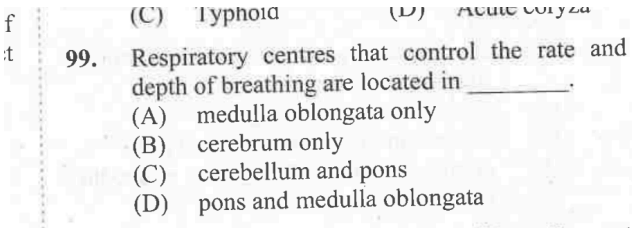


(D) i-d ii-a iii-b iv-c

98. Which one of the following diseases is caused by a fungus?
(A) Filariasis (B) Ringworm
(C) Typhoid (D) Acute coryza

Which one of the following diseases is caused by a fungus?
(A) Filariasis
(B) Ringworm
(C) Typhoid
(D) Acute coryza

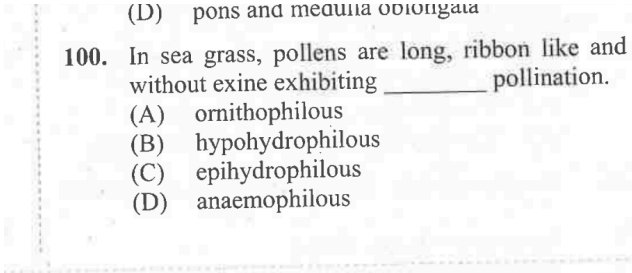
Bio_q099 key **D**


(C) Typhoid (D) Acute coryza

99. Respiratory centres that control the rate and depth of breathing are located in _____.
(A) medulla oblongata only
(B) cerebrum only
(C) cerebellum and pons
(D) pons and medulla oblongata

Respiratory centres that control the rate and depth of breathing are located in _____.
(A) medulla oblongata only
(B) cerebrum only
(C) cerebellum and pons
(D) pons and medulla oblongata

Bio_q100 key **B**


(D) pons and medulla oblongata

100. In sea grass, pollens are long, ribbon like and without exine exhibiting _____ pollination.
(A) ornithophilous
(B) hypohydrophilous
(C) epihydrophilous
(D) anaemophilous

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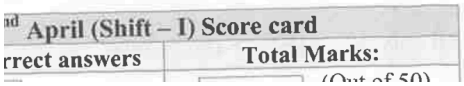
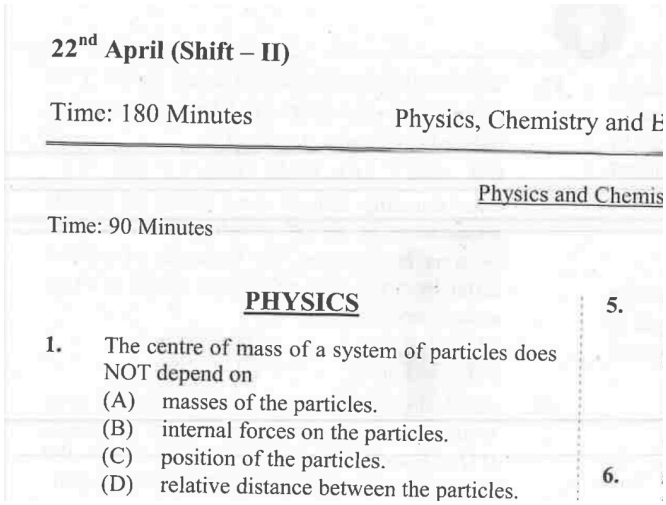

April (Shift – I) Score card
Total Marks:


22nd April (Shift – II)

Time: 180 Minutes Physics, Chemistry and E

Physics and Chemis

Time: 90 Minutes

PHYSICS

1. The centre of mass of a system of particles does NOT depend on
(A) masses of the particles.
(B) internal forces on the particles.
(C) position of the particles.
(D) relative distance between the particles.

5.

6.